COMPREHENSIVE EXAMINATIONS - MCQS CN

Question 1

What is the main purpose of public key cryptography?

a) Encryption

b) Authentication

c) Key exchange

d) All of the above

Question 2

Which algorithm is commonly used for creating and verifying digital signatures?

a) RSA

b) AES

c) DES

d) SHA

Question 3

What is authentication in cryptography?

a) The process of encrypting a message

b) The process of verifying the identity of a user

c) The process of generating a digital signature

d) The process of decrypting a message

Question 4

Which of the following is an example of something you know for authentication?

a) Password

b) Fingerprint

c) Smart card

d) All of the above

Question 5

Which type of authentication uses multiple factors

a) Single-factor authentication

b) Two-factor authentication

c) Three-factor authentication

d) Multi-factor authentication

Question 6

Which type of authentication uses a password and a security token?

a) Single-factor authentication

b) Two-factor authentication

c) Three-factor authentication

d) Multi-factor authentication

Question 7

Which type of authentication uses biometric data?

a) Single-factor authentication

b) Two- factor authentication

c) Three-factor authentication

d) Multi-factor authentication

Question 8

Which type of authentication uses a smart card and a PIN?

a) Single-factor authentication

b) Two-factor authentication

c) Three-factor authentication

d) Multi-factor authentication

Question 9

Which type of attack involves guessing passwords until the correct one is found?

a) Brute force attack

b) Dictionary attack

c) Social engineering attack

d) Man-in-the-middle attack

Question 10

Which type of attack involves using pre-computed hashes to guess passwords?

a) Brute force attack

b) Dictionary attack

c) Rainbow table attack

d) Man-in-the-middle attack

Question 11

Which type of attack involves tricking someone into revealing their password?

a) Brute force attack

b) Dictionary attack

c) Social engineering attack

d) Man-in-the-middle attack

Question 12

Which key is used for encryption in a public key cryptosystem?

a) Public key

b) Private key

c) Symmetric key

d) None of the above

Question 13

Which type of attack involves intercepting and modifying communication between two
 parties?

a) Brute force attack

b) Dictionary attack

c) Social engineering attack

d) Man-in-the- middle attack

Question 14

Which type of encryption algorithm uses the same key for encryption and decryption?

a) Public key encryption algorithm

b) Private key encryption algorithm

c) Symmetric key encryption algorithm

d) Asymmetric key encryption algorithm

Question 15

Which type of encryption algorithm uses different keys for encryption and decryption?

a) a.Public key encryption algorithm

b) b. Private key encryption algorithm

c) c. Symmetric key encryption algorithm

d) d. Asymmetric key encryption algorithm

Question 16

Which type of encryption algorithm is faster?

a) Public key encryption algorithm

b) Private key encryption algorithm

c) Symmetric key encryption algorithm

d) Asymmetric key encryption algorithm

Question 17

Which type of encryption algorithm is more secure?

a) Public key encryption algorithm

b) Private key encryption algorithm

c) Symmetric key encryption algorithm

d) Asymmetric key encryption algorithm

Question 18

Which protocol is used for secure communication over the internet?

a) TLS/SSL

b) HTTPS

c) SSH

d) TCP/IP

Question 19

Which protocol is used for secure remote login and other secure network services over an
 insecure network?

a) TLS/SSL

b) HTTPS

c) SSH

d) TCP/IP

Question 20

Anarkali digitally signs a message and sends it to Salim. Verification of the signature by Salim requires

a) Anarkali’s Public Key

b) Salim’s Public Key

c) Salim’s Private Key

d) Anarkali’s Private Key

Question 21

A sender S sends a message m to receiver R, which is digitally signed by S with its private key. In this scenario, one or more of the following security violations can take place.
 (I) S can launch a birthday attack to replace m with a fraudulent message.
 (II) A third party attacker can launch a birthday attack to replace m with a fraudulent message.
 (III) R can launch a birthday attack to replace m with a fraudulent message.
 Which of the following are possible security violations.

a) (I) and (II) only

b) (I) only

c) (II) only

d) (II) and (III) only

Question 22

AES uses a _____ bit block size and a key size of _____ bits

a) 128; 128 or 256

b) 64; 128 or 192

c) 256; 128, 192, or 256

d) 128; 128, 192, or 256

Question 23

Which key is kept secret in a public key cryptosystem?

a) Public key

b) Private key

c) Symmetric key

d) None of the above

Question 24

How many rounds does the AES-256 perform?

a) 10

b) 12

c) 14

d) 16

Question 25

From the following options, what is the key size in the S-AES algorithm?

a) 16 bits

b) 32 bits

c) 24 bits

d) None of the above

Question 26

Among the fourth-generation firewalls, Dynamic packet filters firewall that work at_____.

a) Application Layer

b) TCP

c) UDP

d) TCP, UDP

Question 27

Authentication and Access control for resources are provided by Network security

a) True

b) False

Question 28

. _____ is the process of verifying the identity of a user.

a) Authentication

b) Identification

c) Validation

d) Verification

Question 29

A digital signature needs a

a) Private-key system

b) Shared-key system

c) Public-key system

d) All of them

Question 30

Encryption and decryption provide secrecy, or confidentiality, but not.....

a) Authentication

b) Integrity

c) Privacy

d) All of the above

Question 31

Decrypt the following ciphertext by using Ceaser Cipher
 HQFUBSWHG WHAW

a) ABANDONED LOCK

b) ENCRYPTED TEXT

c) ABANDONED TEXT

d) ENCRYPTED LOCK

Question 32

A _____ signature is included in the document; a _____
 signature is a separate entity

a) conventional; digital

b) digital; digital

c) either (a) or (b)

d) neither (a) nor (b)

Question 33

Digital signature provides

a) authentication

b) nonrepudiation

c) both (a) and (b)

d) neither (a) nor (b)

Question 34

Which algorithm is commonly used for generating public and private keys?

a) RSA

b) AES

c) DES

d) SHA

Question 35

Digital signature cannot provide _____ for the message

a) integrity

b) confidentiality

c) nonrepudiation

d) . authentication

Question 36

If _____ is needed, a cryptosystem must be applied over the scheme.

a) integrity

b) confidentiality

c) nonrepudiation

d) authentication

Question 37

The secret key betweenmembers needs to be created as a _____key when two members contact KDC.

a) public

b) session

c) complimentary

d) none of the above

Question 38

. _____ is a popularsession key creator protocol thatrequires an authentication
 server and a ticket-granting server.

a) KDC

b) Kerberos

c) CA

d) none of the above

Question 39

CHAP stands for

a) Challenge Handshake authentication protocol

b) Challenge Hardware authentication protocol

c) Circuit Hardware authentication protocol

d) Circuit Handshake authentication protocol

Question 40

SSL provides _____.

a) message integrity

b) confidentiality

c) Both A and B

d) None of the above.

Question 41

Which one of the following is not a public key distribution means?

a) Public-Key Certificates

b) Hashing Certificates

c) Publicly available directories

d) Public-Key authority

Question 42

Which of the following public key distribution systems is most secure?

a) Public-Key Certificates

b) Public announcements

c) Publicly available directories

d) Public-Key authority

Question 43

Which system uses a trusted third party interface?

a) Public-Key Certificates

b) Public announcements

c) Publicly available directories

d) Public-Key authority

Question 44

The subject unique identifier of the X.509 certificates was added in which version?

a) 1

b) 2

c) 3

d) 4

Question 45

What is the primary advantage of using public key cryptography?

a) Faster encryption speed

b) Simpler key management

c) Stronger security

d) Lower computational overhead

Question 46

A total of seven messages are required in the Public-Key distribution scenario. However, the initial five messages need to be used only infrequently because both A and B can save the other's public key for future – a technique known as ____

a) time stamping

b) polling

c) caching

d) squeezing

Question 47

In computing, _____ is a network security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules

a) Cookie

b) Spyware

c) Firewall

d) Spam

Question 48

A digital signature is required for _____

a) non-repudiation of communication by a sender

b) all e-mail sending

c) all DHCP server

d) FTP transactions

Question 49

To authenticate thedata origin, one needs a

a) MDC

b) MAC

c) either (a) or (b)

d) neither (a) nor (b)

Question 50

A layer-4 firewall (a device that can look at all protocol headers up to the transport layer) cannot block _____

a) entire HTTP traffic during 9.00PM and 5.00 AM.

b) all ICMP traffic.

c) incoming traffic from a specific IP address but allow outgoing traffic to the same IP address

d) TCP traffic from a specific user on a multi-user system during 9.00 PM and 5.00 AM.

Question 51

A session symmetric key between two parties is used

a) Only once

b) Twice

c) Multiple times

d) Conditions dependant

Question 52

. One security protocol for the e-mail system is _____.

a) IPSec

b) SSL

c) PGP

d) none of the above.

Question 53

. _____ operates in the transport mode or the tunnel mode.

a) IPSec

b) SSL

c) PGP

d) none of the above

Question 54

_____ provides authentication at the IP level

a) AH

b) ESP

c) PGP

d) SSL

Question 55

What is a digital signature?

a) A message that is encrypted with a private key

b) A message that is encrypted with a public key

c) A message that is hashed with a private key

d) A message that is hashed with a public key

Question 56

What is the main purpose of a digital signature?

a) Encryption

b) Authentication

c) Key exchange

d) All of the above

Question 57

Which key is used for creating a digital signature?

a) Public key

b) Private key

c) Symmetric key

d) None of the above

Question 58

Which key is used for verifying a digital signature?

a) Public key

b) Private key

c) Symmetric key

d) None of the above

Question 59

_____ error means that two or more bits in the data unit have changed

a) Burst

b) Double Bit

c) Single Bit

d) None

Question 60

Which Topology is used in Ethernet?

a) Bus

b) Star

c) Mesh

d) Ring

Question 61

A signal ranges from 1 KHz to 40 MHz? Find out the bandwidth.

a) 39.999 MHz

b) 3990 Hz

c) 3 KHz

d) 39 MHz

Question 62

A computer is connected directly to the internet, so what type of address it must have?

a) Public Address

b) Multicast Address

c) Private Address

d) None of the options

Question 63

What is the difference between a switch and a hub?

a) Hub transmit a signal to all the devices connected to it, whereas switch transmits a signal to four intended ports

b) Switch works in physical layer, hub works at data-link layer

c) Switch works at layer 2 while hub works at layer 1

d) Hub is a smart device, whereas switch is a dumb device

Question 64

Assume 10 devices are arranged in a mesh topology. How many cables are needed? How many ports are needed for each device?

a) 90,9

b) 110,9

c) 45,9

d) 55,9

Question 65

Which of the following are not the functions of router
 i. Data conversion for format to another
 ii. Forward the packets
 iii. Constructs the routing table
 iv. Detects error in the packet
 v. Queue the incoming packets
 vi. Forward the packet to all outgoing link

a) i, iv, vi

b) i, v, vi

c) i, ii, v, vi

d) iv, v, vi

Question 66

What happens during the failure of a router in the virtual switching circuit?

a) All the crashed packets get lost

b) All the virtual circuits that passes through the router get terminated

c) Congestion control becomes extremely difficult

d) Sequence of packets arrival gets disrupted

Question 67

How many cross points are needed in a single stage switch with 30 inputs and 30 outputs.

a) 20

b) 30

c) 900

d) 600

Question 68

Which of the following function is not used by TCP server

a) Connect()

b) Socket()

c) Bind()

d) Listen()

Question 69

Flow control is mainly a function of the layer

a) Application

b) Presentation

c) Session

d) Data Link

Question 70

What is the hamming distance between 10101010 and 01010101

a) 8

b) 0

c) 4

d) none

Question 71

What is network address for 12.12.12.12/20

a) 12.12.1.1

b) 12.12.0.2

c) 12.12.12.0

d) 12.12.0.0

Question 72

Which routing protocol allows throughput to be used as a cost variable for taking routing decision.

a) RIP

b) Distance Vector routinhg

c) OSPF

d) Interadomain

Question 73

In the IPv6 header, the traffic class field is similar to which field in the IPv4 header?

a) Fragmentation Field

b) Fast - Switching

c) ToS Field

d) Option Field

Question 74

Identify the class of the following IPv4 binary address: 11011110 10101001 10101010 10011001.

a) Class A

b) Class B

c) Class C

d) Class D

Question 75

_____Layer do the task of aggregating the data from different processes and pass that data to _____Layer

a) Application, Session

b) Presentation, Session

c) Transport, Network

d) Network, Data Link

Question 76

_____ are the Application Layer Protocol

a) HTTP, SMTP, SNMP

b) FTP, ICMP, IGMP

c) DHCP, ARP, ICMP

d) None of the options

Question 77

In the sliding window method of flow control, if 100 good data frames are sent and received, how many ACK frames have been sent by the receiver?

a) 99

b) 100

c) 101

d) Some number less than or equal to 100

Question 78

Checksum uses _____ arithmetic

a) 1's Complement

b) 2's Complement

c) None of the options

d) None

Question 79

A socket uses an address to establish the connection with some other socket. This socket address is a combination of _____and _____.

a) IP address and Port number

b) MAC address Port number

c) Port Number and Application ID

d) Email address and IP address

Question 80

While using a TCP connection for communication at an instance receiver window size is 5KB and the congestion window size is 2KB. What should be the size of the sending window?

a) 3KB

b) 2KB

c) 4KB

d) 5KB

Question 81

A receiver receives the code 11001100111. When it uses the Hamming encoding algorithm, the result is 0101. Which bit is in error?

a) 1

b) 3

c) 5

d) No Error

Question 82

Network layer add its own header and forward the packet to next layer. In a typical scenario optional information of 20 bytes is added in the header. What is the size of header?

a) 10

b) 20

c) 30

d) 40

Question 83

Adding 1 and 1 in modulo-2 arithmetic results in _____

a) 0

b) 1

c) 2

d) 11

Question 84

The Completion of one full pattern of a signal is called a

a) Period

b) Cycle

c) Frame

d) Segment

Question 85

FTP follows a _____approach to transfer data between a client and a server.

a) Client and Server

b) Peer to Peer

c) Cloud based

d) Parallel

Question 86

Consider two nodes A and B are communicating using FTP. Ten files need to be transferred from A to B. How many connections are required for this purpose?

a) 10

b) 11

c) 12

d) 13

Question 87

Which of the protocol can be used by the receiver to extract the email from its server?

a) SMTP

b) IMAP

c) FTP

d) TCP

Question 88

The term that refers to infinite number of values in the range is

a) Peak

b) Analog Signal

c) Digital Signal

d) None of the options

Question 89

Both Analog and Digital signals can be

a) Periodic

b) Non-Periodic

c) Periodic or Non-Periodic

d) None of the options

Question 90

To efficiently manage the hardware and software components configuration management is required. It includes:

a) Monitoring the user behavior and his/her usage pattern

b) Replacement of the network components

c) Documentation is done only for the hardware changes in the network, software update is not required.

d) Managing the access of the user's to the network.

Question 91

Pick the Odd one of the following as the function of Network Layer?

a) Routing

b) Inter-Networking

c) Error Control

d) None of the options

Question 92

In a _____ network, two types of addressing are involved: global and local

a) Virtual Circuit

b) Datagram

c) Circuit-Switched

d) All the options

Question 93

In which switching store and forward is used?

a) Packet Switching

b) Circuit Switching

c) Cell Switching

d) None

Question 94

Circuit switching is employed in

a) Physical layer

b) Datalink layer

c) Network layer

d) None

Question 95

Packet switching is employed in

a) Physical layer

b) Datalink layer

c) Network layer

d) None

Question 96

Which one of the techniques belongs to connection oriented?

a) Packet Switching

b) Circuit Switching

c) Router

d) None

Question 97

Which one of the techniques belongs to connectionless?

a) Packet Switching

b) Circuit Switching

c) Router

d) None

Question 98

Which switching does need to establish connection and provides dedicated connection between hosts?

a) Packet switching

b) Circuit switching

c) Cell switching

d) None

Question 99

Mesh topology leads to

a) Redundant connections

b) Single point of failure

c) Hard to add new devices

d) None

Question 100

Star topology is described as

a) Every device in the network is connected to all other devices

b) Every device in the network is connected to two other nodes

c) Every device in the network is individually connected to a central node

d) None

Question 101

Which one does support error or flow control?

a) Frame Relay

b) TCP

c) UDP

d) None

Question 102

Which one is a connection oriented data service?

a) AAL 2

b) AAL 3

c) AAL 4

d) None

Question 103

In an organization with its own private network, most of the applications transmit at constant rates for long duration all the time. Which type of switching is preferred?

a) Packet

b) Circuit

c) Datagram

d) All the options

Question 104

Which one is not present in TCP/IP model?

a) Network Layer

b) Transport Layer

c) Session Layer

d) None

Question 105

Which layer that determines the speed of the data transmission in the network?

a) Logical Layer

b) Physical Layer

c) Datalink Layer

d) None

Question 106

Encryption and Decryption is done by

a) Session layer

b) Application layer

c) Presentation layer

d) None

Question 107

Which one is an appropriate example of Half Duplex?

a) Cardless phone

b) Telephone

c) Walkie Talkie

d) None

Question 108

Coverage area of the LAN technology is

a) few kilometers

b) 100 – 1000 kilometers

c) 5 to 50 kilometers

d) None

Question 109

Data transmission speed of LAN technology is

a) upto 100 Mbps

b) upto 20 Mbps

c) upto 1000 Mbps

d) None

Question 110

The functionality of the ARP is

a) to map a logical address with a physical address

b) to map a physical address with a logical address

c) to map virtual address to logical address

d) None

Question 111

Which one is the MAC address?

a) 192.168.10.2

b) 07:01:02:01 :2C:4B

c) 192:220:02:01 :2C

d) None

Question 112

What is the size of IPV4?

a) 64 bits

b) 128 bits

c) 32 bits

d) none

Question 113

Which one allows for communication between networks?

a) Switch

b) Router

c) Repeater

d) None

Question 114

In the _____ approach of packet switching, all packets of a message follow the same route from sender to receiver.

a) Datagram

b) Virtual circuit

c) Virtual channel

d) None of the options

Question 115

Which one allows for communication between two different networks?

a) Switch

b) Router

c) Gateway

d) None

Question 116

The data unit 10100101 is changed as 10100100 referred as

a) Single bit error

b) Burst error

c) Common error

d) None

Question 117

What is the hamming distance between the pair of words (10101,11110)?

a) 2

b) 3

c) 4

d) None

Question 118

Simple parity check can detect

a) An odd number of errors

b) An even number of errors

c) Both odd and even number of errors

d) None of the options

Question 119

Static routing table

a) Contains information entered manually

b) is updated periodically by using one of the dynamic routing protocols

c) is updated periodically by using one of the dynamic routing protocols

d) None

Question 120

Which one is intra domain routing protocols?

a) Distance vector

b) Path vector

c) Border gateway routing protocol

d) None

Question 121

Which one is inter domain routing protocols?

a) Distance vector

b) Path vector

c) Link state routing protocol

d) None of the options

Question 122

Routing Information Protocol is an

a) Intradomain routing protocol based on distance vector routing

b) Intradomain routing protocol based on link state routing

c) interdomain routing protocol based on distance vector routing

d) None

Question 123

Path vector routing protocol can be used

a) only inside an autonomous system

b) between autonomous systems

c) in intradomain

d) None

Question 124

Broadcast has the relationship between the source and the destination is

a) one-to-all

b) one-to-many

c) one-to-one

d) None

Question 125

A Switched network consists of a series of interlinked nodes, called

a) Endpoints

b) Packets

c) Switches

d) Lines

Question 126

Bursty data is

a) the rate of the data flow changes in time, with the changes smooth instead of sudden and sharp

b) the data rate that does not change

c) the data rate changes suddenly in a very short time

d) None

Question 127

Discarding policy that

a) routers may prevent congestion and at the same time may not harm the integrity of the transmission

b) can also prevent congestion in virtual-circuit networks

c) If the sender feels that a sent packet is lost or corrupted, the packet needs to be retransmitted

d) None

Question 128

Node to node congestion control is called as

a) Choke packet

b) Back Pressure

c) Front Pressure

d) None

Question 129

The data must arrive at the receiver exactly as they were sent is referred as

a) Message confidentiality

b) Message Integrity

c) Message Authentication

d) None

Question 130

Which one is the type of relationship between signature and documents in Digital Signature?

a) One to many

b) Many to one

c) One to one

d) None

Question 131

The Topology with highest reliability is

a) Star

b) Bus

c) Mesh

d) Ring

Question 132

Internet is a type of

a) Wide area Network

b) Local area Network

c) Personal area network

d) Metropolitan area network

DSA

Question 1

If the height of the binary tree is 7, maximum number of internal nodes the tree can have is,

a) 128

b) 127

c) 64

d) 63

Question 2

The following is the result after first partition of quicksort algorithm 10,8, 6, 9, 11, 14, 16, 15. Which of the following is true?

a) 11 could be the pivot , 14 could not be pivot.

b) 14 could be the pivot , 11 could not be pivot.

c) Either 11 or 14 could be pivot element

d) 11 and 14 both are pivot elements.

Question 3

What is the advantage of recursive approach than an iterative approach?

a) Consumes less memory

b) Less code and easy to implement

c) Consumes more memory

d) All the options

Question 4

Consider the array LA = {1, 2, 4, 3} is sorted using bubble sort. How many iterations is required in order to sort the array?

a) 4

b) 2

c) 1

d) 0

Question 5

Consider the following code snippet
 char c[]=”VITEEE2022”;
 char *p=c;
 printf(“%s”,p+p[1]-p[3]);

a) EE2022

b) EEE2022

c) E2022

d) Error

Question 6

int main()
 {
 int s[5];
 //considering the initial memory location of s is 1000 and one integer is of 4 byte
 s++;
 printf(“the array is at memory location %d”,s);
 return 0;
 }
 Predict the output

a) 1002

b) 1004

c) 1020

d) Error

Question 7

Run time of the recurrence T(n) = 7T(n/3)+n^2 log n.

a) Theta(n^2 log n)

b) Theta (n^2)

c) Theta(n^2 Log^2 n

d) Theta(n log^2 n)

Question 8

Minimum (Asymptotic) time required to check whether a given binary tree is a binary search tree or not.

a) O(1)

b) O(n)

c) O(log n)

d) O(n log n)

Question 9

If you keep track of front, rear and count(no. of items in array), in a circular array when front =0 and rear=maxsize-1, what will be the count value.

a) 0

b) 3

c) 4

d) 2

Question 10

Which of the following data structure can be applied directly on scheduling and resource allocation problems?

a) stack

b) queue

c) tree

d) linked list

Question 11

If you are asked to implement stack using queue how many queues are required to represent stack efficiently.

a) 1

b) 3

c) 4

d) 2

Question 12

Which is something you would NOT use a stack for?
 i. Printing a name in reverse order
 ii. A function call within another function call
 iii. Arranging the elements of an array in increasing or decreasing order.
 iv. Printing the elements of an array in the order they appear within the array.

a) only (i)

b) only (ii) and (iii) are correct

c) only (iii) and (iv) are correct

d) None of the options are correct

Question 13

Which of the following type of queue is useful in building online multiplayer games.

a) Simple queue

b) Circular Queue

c) Priority Queue

d) Single Ended Queue

Question 14

In a linked list, there is a pointer to the first node, each node has pointer to the next node and last node has pointer to the first node, the name of the linkedlist is,

a) Singly linkedlist

b) Doubly linkedlist

c) Singly Circular linkedlist

d) Doubly Circular linkedlist

Question 15

An array A of integers is given as A: array [1 ... 5] [1 ... 10]; The array is stored in column-major order and the first element of .The array is stored at location 50, Calculate the address of the element A[i][j] where each integer takes one memory location.

a) 5i + j + 44

b) i + 5j + 44

c) 5i + j + 39

d) 5j + i + 39

Question 16

In a binary tree, which of the following is not a valid argument?

a) It should have at least one node.

b) 2

c) 3

d) 0

Question 17

We are given a stack on which two operations are possible, Push( ) and Pop( ). Further, we are given an input sequence 7, 5, 2, 1, 3, 6, where 7 is the first element and 6 is the last element. Which sequence of elements cannot be printed using the given stack?

a) 6 3 1 2 5 7

b) 2 1 3 6 7 5

c) 5 2 7 1 3 6

d) None of the options

Question 18

Which of the given options provides the decreasing order of asymptotic complexity of functions f1, f2, f3 and f4?
 f1=O(n),f2=O(log n), f3=O(n^2) f4=O(2n)

a) f1, f2, f3, f4

b) f2, f1, f4,f3

c) f1, f2, f4, f3

d) f1, f3, f2, f4

Question 19

Which of the following methods have time complexity O(n) for a sorted array in descending order, where elements are distinct?
 (i) Finding maximum element in the array.
 (ii) Finding minimum element in the array.
 (iii) Finding the mode of the array.
 (iv) Find any element in an array.
 Algorithm XY()
 {
  i=1, j=1
  while(i<=n)
 {
  while(j<=n)
  {
  print(“Chennai”)
  j=j+1
  }
  }
 }

a) (i)

b) (ii),(iii),(iv)

c) (iii),(iv)

d) (i),(iii)

Question 20

What is the infix version of the following postfix expression?
 x 12 + z 17 y + 42 * / +

a) x + 12 + z / ((17 + y) * 42)

b) x + 12 + z / 17 + y * 42

c) x + 12 + z / (17 + y) * 42

d) x + (12 + z) / (17 + y * 42)

Question 21

Suppose T1(n) = O(F (n)) and T2(n) = O(F (n)). Which of the following are true?

a) T1(n) + T2(n) = O(F (n))

b) T1(n) *T2(n) = O(F (n))

c) T1(n)=T2(n) = O(1)

d) T1(n) = O(T2(n))

Question 22

In a connected graph with no loops or multiple edges, which of the following inequalities is not correct? (v is the number of vertices, e is the number of edges)

a) $e \leq v \wedge 2$

b) $e \geq v - 1$

c) $v \geq e/2$

d) All the options

Question 23

A circular doubly linked list has exactly two nodes, each of whose addresses are stored in two pointer variables, say head and tail. Which among the following statements is NOT True?

a) Head->next=tail

b) Head->prev->next=tail

c) Tail->prev->next=tail

d) Tail->next->prev=tail

Question 24

We have two stacks, named as A and B. Function Push(&A, data) pushes data on stack A, whereas function Push(&B, data) pushes data on stack B. Pop function on stack A pop out the element from stack A, and pushes it on stack B. On the other hand, Pop function on stack B pops out the element from stack B, and prints it on the screen. Which sequence of character will be printed on the screen after following set of operations: Push(&A, 8), Push(&B, 9), Push(&A, 4), Push(& B, 7), Push(&A,3), Pop(&A), Push(&B, 5), Pop(&A), Pop(&A), Pop(&B), Pop(&B), Pop(&B), Pop(&B), Pop(&B), Pop(&B)?

a) 9 7 3 5 4 8

b) 8 4 5 3 7 9

c) 3 4 8 5 7 9

d) None of the options

Question 25

Which of the following will occur after the following set of operations on an empty stack of size 4. Push(2), Push (3), Pop( ), Push (3), Pop ( ), Pop( ), Push(2), Pop ( ), Pop ( ) ?

a) Stack underflow

b) Stack overflow

c) Operations will not lead to any error.

d) None of the options

Question 26

For the two functions f(n)=n^2+3n+4 and g(n)=n^3 we can conclude that,

a) f(n) = O(g(n))

b) f(n) = Θ(g(n))

c) g(n)=O(f(n))

d) g(n)= Θ(f(n))

Question 27

In a stack 'S' the elements 4, 8, 12, 16 are pushed in the same order. Two times pop operation is executed, sum of the returned values is pushed again in stack S. Again two times pop operation is executed and sum of the returned values are pushed in stack S. The current top of the stack S is

a) 4

b) 16

c) 36

d) 28

Question 28

How many vertices are there in a complete graph with 'n' vertices?

a) n*(n-1)/2

b) n*(n+1)/2

c) 2n

d) n/2

Question 29

What is the value of the sum of the minimum in-degree and maximum out-degree of an Directed Acyclic Graph?

a) Will always be zero

b) Depends on a graph

c) Will always be greater than zero

d) None of the options

Question 30

Traversal of a graph is different than tree because

a) There can be a loop in the graph

b) DFS on a graph uses stack, while inorder traversal is recursive

c) Both the given properties

d) None of the options

Question 31

Why graph traversal is difficult than tree traversal ?

a) Because tree is binary

b) Because tree is undefined

c) Because tree have the root

d) All the options

Question 32

The worst case time complexity of insertion sort is O(n^2). What will be the worst case time complexity of insertion sort if the correct position for inserting element is calculated using binary search?

a) O(n log n)

b) O(n)

c) O(n^2)

d) O(1)

Question 33

Consider the code given below:
 void insertionsort(int arr[],int size)
 {
 int l,j,value;
 for(i=1;i<size;i++)
 {
 value=arr[i];
 j=i;
 while(...)
 {
 arr[j]=arr[j-1];
 j=j-1;
 }
 arr[j]=value;
 }
 }
 what is line of code required for while statement to run the insertion sort

a) (j>0)||(arr[j-1]>value)

b) (j>0)&&(arr[j-1]>value)

c) (j>0)&&(arr[j+1]>value)

d) (j>0)&&(arr[j+1]<value)

Question 34

Consider an array of length 5, arr[5] = {9, 7, 4, 2, 1}. What are the steps of insertions done while running insertion sort on the array?

a) i)7 9 4 2 1
 ii)4 7 9 2 1
 iii) 2 4 7 9 1
 iv) 1 2 4 7 9

b) i) 9 7 4 1 2
 ii) 9 7 1 2 4
 iii) 9 1 2 4 7
 iv) 1 2 4 7 9

c) i) 7 4 2 1 9
 ii) 4 2 1 9 7
 iii) 2 1 9 7 4
 iv) 1 9 7 4 2

d) i) 7 9 4 2 1
 ii) 2 4 7 9 1
 iii) 4 7 9 2 1
 iv) 1 2 4 7 9

Question 35

A connected planar graph having 6 vertices, 7 edges contains _____ regions.

a) 11

b) 15

c) 1

d) 3

Question 36

When the entries 7, 4, 6, 1, 2, 3, 8, 5 are successively inserted into an initially empty binary search tree, what is the height of the resulting tree?

a) 4

b) 3

c) 7

d) 2

Question 37

How many different trees are possible with 10 nodes?

a) 512

b) 1024

c) 1014

d) 90

Question 38

To implement Stacks using Queues, what are the minimum number of queues required?

a) 2

b) 3

c) 1

d) Atleast 3

Question 39

There are 8, 15, 13, 14 nodes in 4 different trees. Which of them could have formed a full binary tree?

a) 8

b) 15

c) 13

d) 14

Question 40

The memory use of an adjacency matrix is

a) O(n)

b) O(n^2)

c) O(n^3)

d) O(logn)

Question 41

A polynomial can be represented in a _____ by just storing the coefficient and exponent of each term.

a) Array

b) Linked list

c) Queue

d) Stack

Question 42

_____ is used to convert an infix expression into a postfix expression.

a) Linked list

b) Stack

c) Queue

d) Tree

Question 43

Which type of linked list does not store NULL in next field?

a) Singly linked list

b) Circular linked list

c) Doubly linked list

d) Linear List

Question 44

The circular queue will be full only when

a) FRONT = MAX –1 and REAR = MAX –1

b) FRONT = 0 and REAR = MAX –1

c) FRONT = MAX –1 and REAR = 0

d) FRONT = 0 and REAR = 0

Question 45

Which type of linked list contains a pointer to the next as well as the previous node in the sequence?

a) Singly linked list

b) Circular linked list

c) Doubly linked list

d) All the options

Question 46

In _____, insertions can be done only at one end, while deletions can be done from both the ends.

a) Double-ended queue

b) Input restricted deque

c) Output restricted deque

d) Circular queue

Question 47

If the values of A, B, C, and D are 2, 3, 4, and 5, respectively, what is the value of the following prefix expressions: a. + – * A B C D

a) 7

b) -3

c) 5

d) 3

Question 48

What does the following lines of C code does?
 CURSOR_SPACE[p].next = CURSOR_SPACE[0].next;
 CURSOR_SPACE[0].next = p;

a) Allocates cursor space

b) Releases cursor space

c) Initialization of cursor space

d) Declaration for cursor

Question 49

If you consider the following as number of operations that different algorithms take to complete a particular task on n elements, which one is linear ?

a) 3n^3+3n + 7

b) 3n^2 + 3n + 2

c) 3n + 1

d) 8

Question 50

Consider the linear arrays AAA(5:50) and BBB(-5:10). The number of elements in each array

a) 50, 18

b) 46, 18

c) 46, 16

d) 50, 16

Question 51

Suppose the contents of an array A are, A = {1, null, null, null, null, 10}. What would be the size of the array considering it as a normal array and a sparse array?

a) 6 and 6

b) 6 and 2

c) 2 and 6

d) 2 and 2

Question 52

Consider the following code.
 int fun()
 {
 if(isEmpty())
 {
 return -10;
 }
 else
 {
 int n;
 n= q[front];
 front++;
 return n;
 }
 }
 Which operation does the above code perform?

a) Enqueue

b) Dequeue

c) Return the front element

d) Dequeue and return the front element

Question 53

In the Deque implementation using singly linked list, what would be the time complexity of deleting an element from the rear end?

a) O(1)

b) O(n^2)

c) O(n)

d) O(nlogn)

Question 54

In a circular queue implementation using an array of size 5, the array index starts with 0 where front and rear values are 3 and 4 respectively. Determine the array index at which the insertion of the next element will take place.

a) 5

b) 0

c) 1

d) 2

Question 55

Which of the following concepts make extensive use of arrays?

a) Binary trees

b) Scheduling of processes

c) Caching

d) Spatial locality

Question 56

Suppose that an intermixed sequence of push and pop operations are performed. The pushes push the integers 0 through 9 in order; the pops print out the return value. Which of the following sequences could not occur?

a) 4 3 2 1 0 9 8 7 6 5

b) 4 6 8 7 5 3 2 9 0 1

c) 2 5 6 7 4 8 9 3 1 0

d) 4 3 2 1 0 5 6 7 8 9

Question 57

Whenever there is a pending operation to be performed, the function becomes _____ recursive.

a) Direct recursive

b) Indirect recursive

c) Tail recursive

d) Non-tail recursive

Question 58

A node is said to be _____ if it has a possibility of reaching a complete solution.

a) Non-promising

b) Promising

c) Succeeding

d) Preceding

Question 59

Which of the following is false in the case of a spanning tree of a graph G?

a) It is tree that spans G

b) It is a subgraph of the G

c) It includes every vertex of the G

d) It can be either cyclic or acyclic

Question 60

In a circular queue with the capacity of n elements, insertion and deletion are performed using rear and front respectively. Initially rear and front are set to 1. Which of the following condition must be true for the queue to be full.

a) front=rear

b) front=rear+1

c) front=rear+1

d) rear=front+1

Question 61

Consider a complete graph G with 4 vertices. The graph G has ____ spanning trees

a) 15

b) 8

c) 16

d) 13

Question 62

A person wants to visit some places. He starts from a vertex and then wants to visit every place connected to this vertex and so on. What algorithm he should use?

a) Depth First Search

b) Breadth First Search

c) Prim’s algorithm

d) Kruskal’s algorithm

Question 63

Which of the following is not an application of Depth First Search?

a) For generating topological sort of a graph

b) For generating Strongly Connected Components of a directed graph

c) Detecting cycles in the graph

d) Peer to Peer Networks

Question 64

Dijkstra's Algorithm cannot be applied on _____

a) Directed and weighted graphs

b) Graphs having negative weight function

c) Unweighted graphs

d) Undirected and unweighted graphs

Question 65

Consider the following code snippet
 char c[]="VITEEE2022";
 char *p=c;
 printf("%s",p+p[1]-p[3]);

a) EE2022

b) EEE2022

c) E2022

d) Error

Question 66

int main()
 {
 int s[5];
 //considering the initial memory location of s is 1000 and one integer is of 4 byte
 s++;
 printf("the array is at memory location %d",s);
 return 0;
 }
 Predict the output

a) 1002

b) 1004

c) 1020

d) Error

Question 67

int main()
 {
 int s[5];
 //considering the initial memory location of s is 1000 and one integer is of 4 byte
 printf(“%d %d”,s+1,&s+1);
 return 0;
 }
 Predict the output

a) 1004 1020

b) 1004 1004

c) 1020 garbage value

d) Not even pass compilation

Question 68

#include<stdio.h>
 int main ()
 {
 int arrayVal[]={5,8,12,14,18,19};
 int *pointerVal=(int*) (&arrayVal +1);
 printf(“%d”,*(pointerVal -1));
 return 0;
 }
 Output of the above will be

a) 12

b) 5

c) 19

d) 18

Question 69

What will be the output of the following code snippet
 int main()
 {
 int i;
 int s[5]={1};
 for(i=0;i<5;i++)
 {
 printf(“%d”,arr[i]);
 }
 return 0;
 }

a) 1 followed by four garbage value

b) 1 0 0 0 0

c) 1 1 1 1 1

d) 0 0 0 0 0

Question 70

Consider the following statement
 int (*p)[5];
 What the above statement represents?

a) Compilation error

b) p is a pointer to 5 integers

c) p is an array of 5 pointers to integers

d) p is a pointer to an array of 5 integer

Question 71

Consider the following pseudocode for the queue Q
 Function(Q)
 If (Q!=empty)
 x=delete(Q)
 Function(Q)
 Insert(Q,x)
 What will be the change in a queue ?

a) Q element order is reversed

b) Q is unchanged

c) Q's Front element is removed and inserted at the rear, other elements are unchanged.

d) Q is emptied

Question 72

What will be the output of the following code snippet
 #inlcude<stdio.h>
 void test(int *ptr)
 {
 ptr=(int*)malloc(sizeof(int));
 }
 int main()
 {
 int *ptr;
 test(ptr);
 *ptr=6;
 printf("%d n",*ptr);
 return(0);
 }

a) May not work

b) Works and 6 will be printed

Question 73

Consider the following code snippet and identify the true statement among the given options
 int i;
 int main()
 {
 int j;
 int *k=(int *)malloc (sizeof(int));
 }

a) i,j and *k are stored in stack segment

b) i and j are stored in stack segment. *k is stored on heap

c) i is stored in static area, j is stored in stack area and *k is stored in heap area.

d) j is stored in static area, i is stored in stack area and *k is stored in heap area.

Question 74

Consider the following code snippet
 #include <stdio.h>
 Int main()
 {
 int num=5;
 int const * p=#
 ++(*p);
 printf(“%d”,num);
 return 0;
 }

a) Compile time error

b) Run time error

c) 6

d) 5

Question 75

Consider the following code snippet
 #include <stdio.h>
 int main()
 {
 int num=5;
 static int num1=num;
 if(num== num1)
 printf(“equal”);
 if(num> num1)
 printf(“greater”);
 else
 printf(“less”);
 return 0;
 }

a) Compile time error

b) Equal

c) Greater

d) Less

Question 76

Consider the following code snippet
 #include <stdio.h>
 int test(int val)
 {
 static int number=1;
 if(val >=5)
 return val;
 val = val +number;
 number++;
 return test(val);
 }
 What will be the value of test(1) will be:

a) 5

b) 6

c) 7

d) 8

Question 77

Consider the following code snippet
 #include <stdio.h>
 int main(void)
 {
 int num=10;
 const int * p=#
 *p=100;
 printf(“i=%d”,num);
 return 0;
 }

a) i=100

b) i=10

c) compile time error

d) Runtime error

Question 78

Part1 (int i) {
 if (i == 0) return;
 printf ("%d", i);
 part2 (i - 2);
 printf ("%d", i);
 }
 Part2 (int i) {
 if (i == 0) return;
 printf ("%d", i;
 part1 (++i);
 printf ("%d", i);
 }
 What will be the output if part1(5) is called

a) 53423122233445

b) 53423120112233

c) 53423122132435

d) 53423120213243

Question 79

Consider the following code snippet
 int part(int n)
 {
 int x = 1, k;
 if (n = = 1) return x;
 for (k =1; k<n; ++k)
 x = x + fun(k) * fun(n–k);
 return x;
 }
 What would be the output for fun(5)

a) 49

b) 50

c) 51

d) 52

Question 80

Consider the following code snippet
 int part(int x)
 {
 int i = 1, n;
 if (x = = 1) return i;
 for (n =1; n<x; ++n)
 i = i + fun(n) * fun(x–n);
 return i;
 }
 What would be the output for fun(5)

a) 49

b) 50

c) 51

d) 52

Question 81

What will be the output of the following code?
 #include <stdio.h>
 #include <string.h>
 int main()
 {
 char str1[] = "VIT", str2[] = " Chennai ";
 //concatenates str1 and str2 and resultant string is stored in str1.
 strcat(str1,str2);
 puts(str1);
 puts(str2);
 return 0;
 }

a) VIT Chennai Chennai

b) VIT

c) VIT Chennai

d) Compilation Error

Question 82

Which of the following statement is more suitable while inserting in a circular queue

a) rear= rear +1

b) rear= rear-1

c) rear= (rear+1) mod SIZE

d) rear=rear mod SIZE +1

Question 83

What will be the output of the C Program?
 #include<stdio.h>
 int main()
 {
 char arr[]=”function\0”;
 int num=strlen(arr);
 printf(“Length of function is %d”,num);
 return 0;
 }

a) 9

b) 7

c) 10

d) 8

Question 84

Consider the following 2-D array initialization. Which of the options is correct?
 int abc[2][] = {1, 2, 3 ,4 };

a) Valid Initalization

b) Invalid Initialization

Question 85

Consider the following 2-D character string
 #define NUMBER_OF_STRING 5
 #define MAX_STRING_SIZE 50
 char arr[NUMBER_OF_STRING][MAX_STRING_SIZE] =
 {"Getting into the Data Structure", "is interesting ", "that make sure to effectively", "organize the data and"“access & apply the operation”
 };
 for (int i = 0; i < NUMBER_OF_STRING; i++)
 {
 printf("'%s' has length %d\n", arr[i], strlen(arr[i]));
 }
 What the above code will print

a) It will print the length of the string

b) It will print all strings in a string array with their lengths

c) It will print all the string in a string array

Question 86

Structure in C language is

a) a collection of elements that can be of same data type

b) a collection of elements that can be of different data type

c) Elements of a structure are called members

d) All the options

Question 87

The function of strncat and strncmp in string.h library is

a) Appends not more than n characters from the string pointed to by source to the end of the string pointed to by destination plus a terminating Null-character and Comparison is limited to the number of characters specified during the function call respectively.

b) Comparison is limited to the number of characters specified during the function call and appends not more than n characters from the string pointed to by source to the end of the string pointed to by destination plus a terminating Null-character respectively.

c) None of the options

Question 88

Consider the following statements about array and structure and choose the correct statements

a) Array is a type of data structure used as container which can hold variables of same type whereas Structure data structure used as container which can hold variables of different types

b) Array it internally implements Pointer which always points to the first element of Array.

c) Structure is a user defined datatype

d) All the statements are true

Question 89

Which of the following is a properly defined struct?

a) struct {int a;}

b) struct a_struct {int a;}

c) struct a_struct int a;

d) struct a_struct {int a;};

Question 90

What is the output of C program with structures.?
 int main()
 {
 structure hotel
 {
 int items;
 char name[10];
 }a;
 strcpy(a.name, "TAJ");
 a.items=10;
 printf("%s", a.name);
 return 0;
 }

a) TAJ

b) Empty string

c) Compiler error

d) None of the options

Question 91

What will be the output of the program in C?
 #include<stdio.h>
 int main()
 {
 ant arr[5], i=0;
 while(i<5)
 arr[i]=++i;
 for(i=0;i<5;i++)
 printf("%d,",arr[i]);
 return 0;
 }

a) 1,2,3,4,5

b) Garbage value, 1, 2, 3, 4,

c) 0, 1, 2, 3, 4,

d) 2, 3, 4, 5, 6,

Question 92

What is the output of this program?
 #include<stdio.h>
 void main()
 {
 int arr[10];
 printf("%d %d",arr[-2],arr[11]);
 }

a) 0 0

b) Garbage value 0

c) Garbage value Garbage value

d) Compilation Error

Question 93

Which of the following recurrence cannot be solved using a master method?

a) T (n) = 2T (n/4) + n ^0.51

b) T (n) = 64T (n/8) – n^2 log n

c) T (n) = √ 2T (n/2) + log n

d) T (n) = 16T (n/4) + n!

Question 94

Let x be an array. Which of the following operations is illegal? (I) ++x (II) x+1 (III) x++ (IV) x*2

a) I and II

b) I, II and III

c) I, III and IV

d) III and IV

Question 95

What is the right way to Initialize array?

a) int num[6] = { 2, 4, 12, 5, 45, 5 };

b) int n{} = { 2, 4, 12, 5, 45, 5 };

c) int n{6} = { 2, 4, 12 };

d) int n(6) = { 2, 4, 12, 5, 45, 5 };

Question 96

How many stacks are required to implement a queue

a) 1

b) 2

c) 3

d) 4

Question 97

#include<stdio.h>
 str[]=”VIT\0Chennai”;
 #include<conio.h>
 void main()
 {
 char printf(“%s”,str);
 }
 What will be the output of the above program

a) VIT Chennai

b) Chennai

c) VIT\0Chennai

d) VIT

Question 98

What is the output of C program with string
 int main()
 {
 char str[]={'V','I','T'};
 printf("%s",str);
 return 0;
 }

a) VIT

b) V

c) VIT\0

d) None of the options

Question 99

What is the output of following C program
 int main()
 {
 char str1[]="VIT";
 char str2[20];
 str2= str1;
 printf("%s",str2);
 return 0;
 }

a) VIT

b) VIT\0

c) Compile Error

d) V

Question 100

what will be the output of the program
 #include<stdio.h>
 #include<string.h>
 void main()
 {
  char str1[20]= “VIT”, char str2[20]= “Chennai”;
 printf(“%s”,strcpy(str2,strcat(str1,str2)));
 }

a) ChennaiVIT

b) Chennai

c) VIT Chennai

d) VIT

Question 101

#include <stdio.h>
 #include <string.h>
 int main()
 {
 char str1[] = "VIT", str2[] = " Chennai ";
 //concatenates str1 and str2 and resultant string is stored in str1.
 strcat(str1,str2);
 puts(str1);
 puts(str2);
 return 0;
 }

a) VIT Chennai Chennai

b) VIT

c) VIT Chennai

d) Compilation Error

Question 102

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a) 90 and 99

b) 90 and 94

c) 89 and 99

d) 89 and 94

Question 103

The number of swapping needed to sort the numbers 5,1,4,3,2 in ascending order, using bubble sort is:

a) 10

b) 9

c) 8

d) 7

Digital MP

Question 1

How many minterms can be formed with two variable inputs

a) Eight minterms

b) Six minterms

c) Five minterms

d) Four minterms

Question 2

What was the JK levels to toggle the switch?

a) J = 0, K = 0

b) J = 1, K = 0

c) J = 0, K = 1

d) J = 1, K = 1

Question 3

How many flip-flops are required to produce a divide-by-128 device?

a) 1

b) 4

c) 6

d) 7

Question 4

The phenomenon of interpreting unwanted signals on J and K while Cp (clock pulse) is HIGH is called _____.

a) parity error checking

b) ones catching

c) digital discrimination

d) digital filtering

Question 5

How the master is enabled in master-slave flip-flop?

a) when the gate is LOW

b) when the gate is HIGH

c) both LOW and HIGH

d) None of the options

Question 6

Which of the following is correct for a gated D flip-flop?

a) The output toggles if one of the inputs is held HIGH.

b) Only one of the inputs can be HIGH at a time.

c) The output complement follows the input when enabled.

d) Q output follows the input D when the enable is HIGH.

Question 7

In a J-K flip-flop if the clock input frequency is 20kHz what is the output at Q

a) constantly LOW

b) constantly HIGH

c) a 20 kHz square wave

d) a 10 kHz square wave

Question 8

If an active-HIGH S-R latch has a 0 on the S input and a 1 on the R input and then the R input goes to 0, the latch will be _____.

a) SET

b) RESET

c) Clear

d) Invalid

Question 9

To made MOD-32 binary counter how many flip-flops are required?

a) 3

b) 45

c) 5

d) 6

Question 10

The reason behind that the delay problems faced in asynchronous counters is eliminated by Synchronous counters :

a) input clock pulses are applied only to the first and last stages

b) input clock pulses are applied only to the last stage

c) input clock pulses are not used to activate any of the counter stages

d) input clock pulses are applied simultaneously to each stage

Question 11

The minimum number of NAND gates required to implement the Boolean function X+XY'+XY'Z is

a) one

b) Two

c) Three

d) None of the options

Question 12

In tabulation method how the procedures is started to specify the function in terms of

a) List of minterms

b) List of maxterms

c) List of integers

d) List of terms

Question 13

The output Y of a Two-bit comparator is logic "1" whenever the two-bit input "A" is greater than the two-bit input "B". The number of combination for which the output is logic "1" is

a) 2

b) 4

c) 6

d) 8

Question 14

Which one of the following set of gates are best suited for 'parity' checking and 'parity' generation.

a) AND, OR, NOT gates

b) EX-NOR or EX-OR gates

c) NAND gates

d) NOR gates

Question 15

Which of the following code is also known as rereflected code

a) Excess-3 codes

b) Gray code

c) BCD code

d) Error code

Question 16

How '-5' is represented in hex format in 2’s complement in 8-bits?

a) (FB)16

b) (7F)16

c) (FF)16

d) (FA)16

Question 17

Which statement below best describes the function of a decoder?

a) A decoder will convert a decimal number into the proper binary equivalent

b) A decoder will convert a binary number into a specific output representing a particular character or digit.

c) Decoders are used to prevent improper operation of digital systems.

d) Decoders are special ICs that are used to make it possible for one brand of computer to talk to another.

Question 18

Which of the following is correct for a gated D-type flip-flop?

a) The Q output is either SET or RESET as soon as the input goes HIGH or LOW

b) The output complement follows the input when enabled

c) Only one of the inputs can be HIGH at a time

d) The output toggles if one of the inputs is held HIGH

Question 19

What labelling code is carried in the K-map

a) 8-4-2-1 binary

b) Hexadecimal

c) Gray

d) Octal

Question 20

A computer has a word length of 8 bits (including sign). It is required to determine the range of integers that can be stored in the computer. If 1's complement is used to represent the numbers, then the range will be

a) -256 to 256

b) -256 to 255

c) -127 to 127

d) -128 to 127

Question 21

Choose the gray code form for the binary value 101011?

a) 101011

b) 110101

c) 011111

d) 111110

Question 22

To design 1-of-32 decoder how many 3-to-8-line decoders are required?

a) 1

b) 2

c) 4

d) 8

Question 23

Consider Binary coded decimal value 0001 0010 0110 . Convert the given value to binary form.

a) 1111110

b) 1111101

c) 1111000

d) 1111111

Question 24

Consider, there are 8 doors in an automated cinema theatre. Minimum, How many number of On/Off switches are required to open any doors using digital circuit?

a) 1

b) 2

c) 3

d) 4

Question 25

What are the output values of the 4-bit comparator when the two input binary numbers are given as A = 1011 and B = 1010.

a) A > B = 1, A < B = 0, A < B = 1

b) A > B = 0, A < B = 1, A = B = 0

c) A > B = 1, A < B = 0, A = B = 0

d) A > B = 0, A < B = 1, A = B = 1

OS

Question 1

What is an operating system?

a) interface between the hardware and application programs

b) collection of programs that manages hardware resources

c) system service provider to the application programs

d) All the options

Question 2

Which one of the following is not a real time operating system?

a) RTLinux

b) Palm OS

c) QNX

d) VxWorks

Question 3

In the non blocking send _____

a) the sending process keeps sending until the message is received

b) the sending process sends the message and resumes operation

c) the sending process keeps sending until it receives a message

d) none

Question 4

In the Zero capacity queue _____

a) the queue can store at least one message

b) the sender blocks until the receiver receives the message

c) the sender keeps sending and the messages don't wait in the queue

d) none

Question 5

The Zero Capacity queue _____

a) is referred to as a message system with buffering

b) is referred to as a message system with no buffering

c) is referred to as a link

d) none

Question 6

Bounded capacity and Unbounded capacity queues are referred to as _____

a) Programmed buffering

b) Automatic buffering

c) User defined buffering

d) No buffering

Question 7

A process that is based on IPC mechanism which executes on different systems and can communicate with other processes using message based communication, is called _____

a) Local Procedure Call

b) Inter Process Communication

c) Remote Procedure Call

d) Remote Machine Invocation

Question 8

The initial program that is run when the computer is powered up is called _____

a) boot program

b) bootloader

c) initializer

d) bootstrap program

Question 9

What is a trap/exception?

a) hardware generated interrupt caused by an error

b) software generated interrupt caused by an error

c) user generated interrupt caused by an error

d) none

Question 10

In a memory mapped input/output _____

a) the CPU uses polling to watch the control bit constantly, looping to see if a device is ready

b) the CPU writes one data byte to the data register and sets a bit in control register to show that a byte is available

c) the CPU receives an interrupt when the device is ready for the next byte

d) the CPU runs a user written code and does accordingly

Question 11

Which module gives control of the CPU to the process selected by the short-term scheduler?

a) dispatcher

b) interrupt

c) scheduler

d) none

Question 12

The processes that are residing in main memory and are ready and waiting to execute are kept on a list called _____

a) job queue

b) ready queue

c) execution queue

d) process queue

Question 13

What does OS X has?

a) monolithic kernel with modules

b) microkernel

c) monolithic kernel

d) hybrid kernel

Question 14

The interval from the time of submission of a process to the time of completion is termed as _____

a) waiting time

b) turnaround time

c) response time

d) throughput

Question 15

Which scheduling algorithm allocates the CPU first to the process that requests the CPU first?

a) first-come, first-served scheduling

b) shortest job scheduling

c) priority scheduling

d) none

Question 16

In priority scheduling algorithm _____

a) CPU is allocated to the process with highest priority

b) CPU is allocated to the process with lowest priority

c) Equal priority processes can not be scheduled

d) None

Question 17

In priority scheduling algorithm, when a process arrives at the ready queue, its priority is compared with the priority of _____

a) all process

b) currently running process

c) parent process

d) init process

Question 18

Which algorithm is defined in Time quantum?

a) shortest job scheduling algorithm

b) round robin scheduling algorithm

c) priority scheduling algorithm

d) multilevel queue scheduling algorithm

Question 19

Process are classified into different groups in _____

a) shortest job scheduling algorithm

b) round robin scheduling algorithm

c) priority scheduling algorithm

d) multilevel queue scheduling algorithm

Question 20

In multilevel feedback scheduling algorithm _____

a) a process can move to a different classified ready queue

b) classification of ready queue is permanent

c) processes are not classified into groups

d) none

Question 21

Which one of the following can not be scheduled by the kernel?

a) kernel level thread

b) user level thread

c) process

d) none

Question 22

CPU scheduling is the basis of _____

a) multiprocessor systems

b) multiprogramming operating systems

c) larger memory sized systems

d) none

Question 23

With multiprogramming _____ is used productively.

a) time

b) space

c) money

d) all

Question 24

In operating system, each process has its own _____

a) open files

b) pending alarms, signals, and signal handlers

c) address space and global variables

d) all

Question 25

What are the two steps of a process execution?

a) I/O & OS Burst

b) CPU & I/O Burst

c) Memory & I/O Burst

d) OS & Memory Burst

Question 26

An I/O bound program will typically have _____

a) a few very short CPU bursts

b) many very short I/O bursts

c) many very short CPU bursts

d) a few very short I/O bursts

Question 27

A process is selected from the _____ queue by the _____ scheduler, to be executed.

a) blocked, short term

b) wait, long term

c) ready, short term

d) a few very short I/O bursts

Question 28

In the following cases non – preemptive scheduling occurs?

a) When a process switches from the running state to the ready state

b) When a process goes from the running state to the waiting state

c) When a process switches from the waiting state to the ready state

d) All

Question 29

The switching of the CPU from one process or thread to another is called _____

a) process switch

b) task switch

c) context switch

d) all

Question 30

What is Dispatch latency?

a) the speed of dispatching a process from running to the ready state

b) the time of dispatching a process from running to ready state and keeping the CPU idle

c) the time to stop one process and start running another one

d) none

Question 31

Scheduling is done so as to _____

a) increase CPU utilization

b) decrease CPU utilization

c) keep the CPU more idle

d) none

Question 32

What is Turnaround time?

a) the total waiting time for a process to finish execution

b) the total time spent in the ready queue

c) the total time spent in the running queue

d) the total time from the completion till the submission of a process

Question 33

Scheduling is done so as to _____

a) increase the turnaround time

b) decrease the turnaround time

c) keep the turnaround time same

d) there is no relation between scheduling and turnaround time

Question 34

What is Waiting time?

a) the total time in the blocked and waiting queues

b) the total time spent in the ready queue

c) the total time spent in the running queue

d) the total time from the completion till the submission of a process

Question 35

In a timeshare operating system, when the time slot assigned to a process is completed, the process switches from the current state to?

a) Suspended state

b) Terminated state

c) Ready state

d) Blocked state

Question 36

Scheduling is done so as to _____

a) increase the waiting time

b) keep the waiting time the same

c) decrease the waiting time

d) none

Question 37

What is Response time?

a) the total time taken from the submission time till the completion time

b) the total time taken from the submission time till the first response is produced

c) the total time taken from submission time till the response is output

d) none

Question 38

Round robin scheduling falls under the category of _____

a) Non-preemptive scheduling

b) Preemptive scheduling

c) All

d) None

Question 39

With round robin scheduling algorithm in a time shared system _____

a) using very large time slices converts it into First come First served scheduling algorithm

b) using very small time slices converts it into First come First served scheduling algorithm

c) using extremely small time slices increases performance

d) using very small time slices converts it into Shortest Job First algorithm

Question 40

The portion of the process scheduler in an operating system that dispatches processes is concerned with _____

a) assigning ready processes to CPU

b) assigning ready processes to waiting queue

c) assigning running processes to blocked queue

d) all

Question 41

What is FIFO algorithm?

a) first executes the job that came in last in the queue

b) first executes the job that came in first in the queue

c) first executes the job that needs minimal processor

d) first executes the job that has maximum processor needs

Question 42

The strategy of making processes that are logically runnable to be temporarily suspended is called _____

a) Non preemptive scheduling

b) Preemptive scheduling

c) Shortest job first

d) First come First served

Question 43

Complex scheduling algorithms _____

a) are very appropriate for very large computers

b) use minimal resources

c) use many resources

d) all

Question 44

What is FIFO algorithm?

a) first executes the job that came in last in the queue

b) first executes the job that came in first in the queue

c) first executes the job that needs minimal processor

d) first executes the job that has maximum processor needs

Question 45

The strategy of making processes that are logically runnable to be temporarily suspended is called _____

a) Non preemptive scheduling

b) Preemptive scheduling

c) Shortest job first

d) First come First served

Question 46

Cascading termination refers to the termination of all child processes if the parent process terminates _____

a) Normally or abnormally

b) Abnormally

c) Normally

d) None

Question 47

There are 10 different processes running on a workstation. Idle processes are waiting for an input event in the input queue. Busy processes are scheduled with the Round-Robin time sharing method. Which out of the following quantum times is the best value for small response times, if the processes have a short runtime, e.g. less than 10ms?

a) tQ = 15ms

b) tQ = 40ms

c) tQ = 45ms

d) tQ = 50ms

Question 48

Orders are processed in the sequence they arrive if _____ rule sequences the jobs.

a) earliest due date

b) slack time remaining

c) first come, first served

d) critical ratio

Question 49

Under multiprogramming, turnaround time for short jobs is usually _____ and that for long jobs is slightly _____

a) Lengthened; Shortened

b) Shortened; Lengthened

c) Shortened; Shortened

d) Shortened; Unchanged

Question 50

Which of the following statements are true?
 I. Shortest remaining time first scheduling may cause starvation
 II. Preemptive scheduling may cause starvation
 III. Round robin is better than FCFS in terms of response time

a) I Only

b) I and III only

c) II and III only

d) All

Question 51

The real difficulty with SJF in short term scheduling is _____

a) it is too good an algorithm

b) knowing the length of the next CPU request

c) it is too complex to understand

d) none

Question 52

The FCFS algorithm is particularly troublesome for _____

a) time sharing systems

b) multiprogramming systems

c) multiprocessor systems

d) operating systems

Question 53

Consider the following set of processes, the length of the CPU burst time given in milliseconds.
 Process Burst time
 P1 6
 P2 8
 P3 7
 P4 3
 Assuming the above process being scheduled with the SJF scheduling algorithm.

a) The waiting time for process P1 is 3ms

b) The waiting time for process P1 is 0ms

c) The waiting time for process P1 is 16ms

d) The waiting time for process P1 is 9ms

Question 54

An SJF algorithm is simply a priority algorithm where the priority is _____

a) the predicted next CPU burst

b) the inverse of the predicted next CPU burst

c) the current CPU burst

d) anything the user wants

Question 55

A solution to the problem of indefinite blockage of low – priority processes is _____

a) Starvation

b) Wait queue

c) Ready queue

d) Aging

Question 56

Which of the following statements are true?
 i) Shortest remaining time first scheduling may cause starvation
 ii) Preemptive scheduling may cause starvation
 iii) Round robin is better than FCFS in terms of response time

a) I only

b) I and III Only

c) II and III only

d) I, II, and III

Question 57

When a process is in a “Blocked” state waiting for some I/O service. When the service is completed, it goes to the _____

a) Terminated state

b) Suspended state

c) Running state

d) Ready state

Question 58

Mutual exclusion implies that _____

a) if a process is executing in its critical section, then no other process must be executing in their critical sections

b) if a process is executing in its critical section, then other processes must be executing in their critical sections

c) if a process is executing in its critical section, then all the resources of the system must be blocked until it finishes execution

d) none

Question 59

Bounded waiting implies that there exists a bound on the number of times a process is allowed to enter its critical section _____

a) after a process has made a request to enter its critical section and before the request is granted

b) when another process is in its critical section

c) before a process has made a request to enter its critical section

d) none

Question 60

A minimum of _____ variable(s) is/are required to be shared between processes to solve the critical section problem.

a) 1

b) 2

c) 3

d) 4

Question 61

The following program consists of 3 concurrent processes and 3 binary semaphores. The semaphores are initialized as S0 = 1, S1 = 0, S2 = 0.
 Process P0
 while(true)
 {
 wait(S0);
 print '0';
 release(S1);
 release(S2);
 }
 Process P1
 wait(S1);
 release(S0);
 Process P2
 wait(S2);
 release(S0);
 How many times will P0 print ‘0’?

a) At least twice

b) Exactly twice

c) Exactly thrice

d) Exactly once

Question 62

Each process Pi, i = 0,1,2,3,.....,9 is coded as follows.
 repeat
 P(mutex)
 {Critical Section}
 V(mutex)
 forever
 The code for P10 is identical except that it uses V(mutex) instead of P(mutex). What is the largest number of processes that can be inside the critical section at any moment (the mutex being initialized to 1)?

a) 1

b) 2

c) 3

d) None

Question 63

Two processes, P1 and P2, need to access a critical section of code. Consider the following synchronization construct used by the processes.
 Process P1 :
 while(true)
 {
 w1 = true;
 while(w2 == true);
 Critical section
 w1 = false;
 }
 Remainder Section
 Process P2 :
 while(true)
 {
 w2 = true;
 while(w1 == true);
 Critical section
 w2 = false;
 }
 Remainder Section
 Here, w1 and w2 have shared variables, which are initialized to false. Which one of the following statements is TRUE about the above construct?

a) It does not ensure mutual exclusion

b) It does not ensure bounded waiting

c) It requires that processes enter the critical section in strict alternation

d) It does not prevent deadlocks but ensures mutual exclusion

Question 64

Semaphore is a/an _____ to solve the critical section problem.

a) after a particular process

b) periodically

c) atomically

d) none

Question 65

What are the two atomic operations permissible on semaphores?

a) wait and signal

b) stop and wait

c) hold and wait

d) none

Question 66

The following pair of processes share a common variable X.
 Process A
 int Y;
 A1: Y = X*2;
 A2: X = Y;
 Process B
 int Z;
 B1: Z = X+1;
 B2: X = Z;

X is set to 5 before either process begins execution. As usual, statements within a process are executed sequentially, but statements in process A may execute in any order with respect to statements in process B.
 How many different values of X are possible after both processes finish executing?

a) 2

b) 3

c) 4

d) 8

Question 67

The program follows to use a shared binary semaphore T.
 Process A
 int Y;
 A1: Y = X*2;
 A2: X = Y;
 signal(T);
 Process B
 int Z;
 B1: wait(T);
 B2: Z = X+1;
 X = Z;
 T is set to 0 before either process begins execution and, as before, X is set to 5.
 Now, how many different values of X are possible after both processes finish executing?

a) 1

b) 2

c) 3

d) 4

Question 68

Transient operating system code is a code that _____

a) stays in the memory always

b) never enters the memory space

c) comes and goes as needed

d) is not easily accessible

Question 69

All processes share a semaphore variable mutex, initialized to 1. Each process must execute wait(mutex) before entering the critical section and signal(mutex) afterward.
 Suppose a

process executes in the following manner.
 signal(mutex);
 .....
 critical section
 .....
 wait(mutex);
 In this situation :

a) a deadlock will occur

b) processes will starve to enter critical section

c) several processes maybe executing in their critical section

d) all

Question 70

All processes share a semaphore variable mutex, initialized to 1. Each process must execute wait(mutex) before entering the critical section and signal(mutex) afterward.
 Suppose a process executes in the following manner.
 wait(mutex);
 .....
 critical section
 .....
 wait(mutex);

a) a deadlock will occur

b) processes will starve to enter critical section

c) several processes maybe executing in their critical section

d) all

Question 71

Consider the methods used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared boolean variables S1 and S2 are randomly assigned.
 Method used by P1 :
 while(S1==S2);
 Critical section
 S1 = S2;
 Method used by P2 :
 while(S1!=S2);
 Critical section
 S2 = not(S1);
 Which of the following statements describes properties achieved?

a) Mutual exclusion but not progress

b) Progress but not mutual exclusion

c) Neither mutual exclusion nor progress

d) Both mutual exclusion and progress

Question 72

What is a reusable resource?

a) that can be used by one process at a time and is not depleted by that use

b) that can be used by more than one process at a time

c) that can be shared between various threads

d) none

Question 73

A system is in the safe state if _____

a) the system can allocate resources to each process in some order and still avoid a deadlock

b) there exist a safe sequence

c) all

d) none

Question 74

The circular wait condition can be prevented by _____

a) defining a linear ordering of resource types

b) using thread

c) using pipes

d) all

Question 75

A problem encountered in multitasking when a process is perpetually denied necessary resources is called _____

a) deadlock

b) starvation

c) inversion

d) aging

Question 76

To avoid deadlock _____

a) there must be a fixed number of resources to allocate

b) resource allocation must be done only once

c) all deadlocked processes must be aborted

d) inversion technique can be used

Question 77

The request and release of resources are _____

a) command line statements

b) interrupts

c) system calls

d) special programs

Question 78

What are Multithreaded programs?

a) lesser prone to deadlocks

b) more prone to deadlocks

c) not at all prone to deadlocks

d) none of the options

Question 79

The portion of the process scheduler in an operating system that dispatches processes is concerned with _____

a) assigning ready processes to waiting queue

b) assigning running processes to blocked queue

c) assigning ready processes to CPU

d) all

Question 80

For Mutual exclusion to prevail in the system _____

a) at least one resource must be held in a non sharable mode

b) the processor must be a uniprocessor rather than a multiprocessor

c) there must be at least one resource in a sharable mode

d) all

Question 81

For a Hold and wait condition to prevail _____

a) A process must be not be holding a resource, but waiting for one to be freed, and then request to acquire it

b) A process must be holding at least one resource and waiting to acquire additional resources that are being held by other processes

c) A process must hold at least one resource and not be waiting to acquire additional resources

d) None

Question 82

Deadlock prevention is a set of methods _____

a) to ensure that at least one of the necessary conditions cannot hold

b) to ensure that all of the necessary conditions do not hold

c) to decide if the requested resources for a process have to be given or not

d) to recover from a deadlock

Question 83

For sharable resources, mutual exclusion _____

a) is required

b) is not required

c) may be or may not be required

d) none

Question 84

The disadvantage of a process being allocated all its resources before beginning its execution is _____

a) Low CPU utilization

b) Low resource utilization

c) Very high resource utilization

d) None

Question 85

To ensure no preemption, if a process is holding some resources and requests another resource that cannot be immediately allocated to it _____

a) then the process waits for the resources be allocated to it

b) the process keeps sending requests until the resource is allocated to it

c) the process resumes execution without the resource being allocated to it

d) then all resources currently being held are preempted

Question 86

Each request requires that the system consider the _____ to decide whether the current request can be satisfied or must wait to avoid a future possible deadlock.

a) resources currently available

b) processes that have previously been in the system

c) resources currently allocated to each process

d) future requests and releases of each process

Question 87

Given a priori information about the _____ number of resources of each type that maybe requested for each process, it is possible to construct an algorithm that ensures that the system will never enter a deadlock state.

a) minimum

b) average

c) maximum

d) approximate

Question 88

A deadlock avoidance algorithm dynamically examines the _____ to ensure that a circular wait condition can never exist.

a) resource allocation state

b) system storage state

c) operating system

d) resources

Question 89

A system is in a safe state only if there exists a _____

a) safe allocation

b) safe resource

c) safe sequence

d) all

Question 90

The FCFS algorithm is particularly troublesome for _____

a) operating systems

b) multiprocessor systems

c) time sharing systems

d) multiprogramming systems

Question 91

A system has 12 magnetic tape drives and 3 processes : P0, P1, and P2. Process P0 requires 10 tape drives, P1 requires 4 and P2 requires 9 tape drives.
 Process
 P0
 P1
 P2
 Maximum needs (process-wise: P0 through P2 top to bottom)
 10
 4
 9
 Currently allocated (process-wise)
 5
 2
 2
 Which of the following sequence is a safe sequence?

a) P0, P1, P2

b) P1, P2, P0

c) P2, P0, P1

d) P1, P0, P2

Question 92

The resource allocation graph is not applicable to a resource allocation system _____

a) with multiple instances of each resource type

b) with a single instance of each resource type

c) single & multiple instances of each resource type

d) none

Question 93

The Banker's algorithm is _____ than the resource allocation graph algorithm.

a) less efficient

b) more efficient

c) equal

d) none

Question 94

A system with 5 processes P0 through P4 and three resource types A, B, C have A with 10 instances, B with 5 instances, and C with 7 instances. At time t0, the following snapshot has been taken:
 Process
 P0
 P1
 P2
 P3
 P4
 Allocation (process-wise : P0 through P4 top TO bottom)
 A B C
 0 1 0
 2 0 0
 3 0 2
 2 1 1
 0 0 2
 MAX (process-wise: P0 through P4 top TO bottom)
 A B C
 7 5 3
 3 2 2
 9 0 2
 2 2 2
 4 3 3
 Available
 A B C
 3 3 2
 The sequence <P1, P3, P4, P2, P0> leads the system to _____

a) an unsafe state

b) a safe state

c) a protected state

d) a deadlock

Question 95

The wait-for graph is a deadlock detection algorithm that is applicable when _____

a) all resources have a single instance

b) all resources have multiple instances

c) all resources have multiple instances

d) all

Question 96

An edge from process Pi to Pj in a wait for graph indicates that _____

a) Pi is waiting for Pj to release a resource that Pi needs

b) Pi is waiting for Pj to release a resource that Pi needs

c) all resources have multiple instances

d) all

Question 97

If the wait for graph contains a cycle _____

a) then a deadlock does not exist

b) then a deadlock exists

c) hen a deadlock exists

d) either deadlock exists or system is in a safe state

Question 98

If deadlocks occur frequently, the detection algorithm must be invoked _____

a) rarely

b) frequently

c) rarely & frequently

d) none

Question 99

For an effective operating system, when to check for deadlock?

a) at fixed time intervals

b) every time a resource request is made

c) every time a resource request is made at fixed time intervals

d) none

Question 100

What is the main function of the command interpreter?

a) to provide the interface between the API and application program

b) to handle the files in the operating system

c) to get and execute the next user-specified command

d) None

Question 101

A deadlock avoidance algorithm dynamically examines the _____ to ensure that a circular wait condition can never exist.

a) operating system

b) resources

c) system storage state

d) resource allocation state

Question 102

Swapping _____ be done when a process has pending I/O, or has to execute I/O operations only into operating system buffers.

a) must never

b) maybe

c) can

d) must

Question 103

The main memory accommodates _____

a) cpu

b) user processes

c) operating system

d) all

Question 104

The operating system is responsible for?

a) bad-block recovery

b) booting from disk

c) disk initialization

d) all

Question 105

The operating system and the other processes are protected from being modified by an already running process because _____

a) every address generated by the CPU is being checked against the relocation and limit registers

b) they have a protection algorithm

c) they are in different memory spaces

d) they are in different logical addresses

Question 106

Using transient code, _____ the size of the operating system during program execution.

a) maintains

b) changes

c) increases

d) decreases

Question 107

The operating system maintains a _____ table that keeps track of how many frames have been allocated, how many are there, and how many are available.

a) memory

b) mapping

c) page

d) frame

Question 108

To obtain better memory utilization, dynamic loading is used. With dynamic loading, a routine is not loaded until it is called. For implementing dynamic loading _____

a) special support from operating system is essential

b) special support from hardware is required

c) user programs can implement dynamic loading without any special support from hardware or operating system

d) special support from both hardware and operating system is essential

Question 109

The _____ presents a uniform device-access interface to the I/O subsystem, much as system calls provide a standard interface between the application and the operating system.

a) Device drivers

b) I/O systems

c) Devices

d) Buses

Question 110

In real time operating system _____

a) process scheduling can be done only once

b) all processes have the same priority

c) kernel is not required

d) a task must be serviced by its deadline period

Question 111

In Operating Systems, which of the following is/are CPU scheduling algorithms?

a) Priority

b) Round Robin

c) Shortest Job First

d) All

Question 112

Hard real time operating system has _____ jitter than a soft real time operating system.

a) equal

b) more

c) less

d) none

Question 113

For real time operating systems, interrupt latency should be _____

a) zero

b) minimal

c) maximum

d) dependent on the scheduling

Question 114

Which one of the following is a real time operating system?

a) Windows CE

b) RTLinux

c) VxWorks

d) All

Question 115

The priority of a process will _____ if the scheduler assigns it a static priority.

a) depends on the operating system

b) change

c) remains unchanged

d) none

Question 116

If the sum of the working set sizes increases, exceeding the total number of available frames _____

a) the operating system selects a process to suspend

b) the system crashes

c) then the process crashes

d) the memory overflows

Question 117

The information about all files is kept in _____

a) operating system

b) separate directory structure

c) swap space

d) none

Question 118

The operating system keeps a small table containing information about all open files called _____

a) file table

b) directory table

c) open-file table

d) system table

Question 119

What will happen in the single level directory?

a) All files are contained in the same directory

b) All files are contained in different directories all at the same level

c) Depends on the operating system

d) None

Question 120

The operating system _____ the links when traversing directory trees, to preserve the acyclic structure of the system.

a) deletes

b) considers

c) ignores

d) none

Question 121

On systems where there are multiple operating system, the decision to load a particular one is done by _____

a) process control block

b) file control block

c) boot loader

d) bootstrap

Question 122

To access the services of the operating system, the interface is provided by the _____

a) Library

b) System Calls

c) Assembly instructions

d) API - Application Programming Interface

Question 123

The _____ program initializes all aspects of the system, from CPU registers to device controllers and the contents of main memory, and then starts the operating system.

a) bootstrap

b) main

c) bootloader

d) rom

Question 124

Network operating system runs on _____

a) every system in the network

b) server

c) both server and every system in the network

d) none

Question 125

What are the types of distributed operating systems?

a) Zone based Operating system

b) Level based Operating system

c) Network Operating system

d) All

Question 126

The systems which allow only one process execution at a time, are called _____

a) uniprogramming systems

b) uniprocessing systems

c) unitasking systems

d) none

Question 127

In operating system, each process has its own _____

a) address space and global variables

b) open files

c) pending alarms, signals and signal handlers

d) all

Question 128

A process can be terminated due to _____

a) normal exit

b) fatal error

c) killed by another process

d) all

Question 129

What is the ready state of a process?

a) when process is scheduled to run after some execution

b) when process is unable to run until some task has been completed

c) when process is using the CPU

d) none

Question 130

What is interprocess communication?

a) communication within the process

b) communication between two process

c) communication between two threads of same process

d) none

Question 131

A set of processes is deadlock if _____

a) each process is blocked and will remain so forever

b) each process is terminated

c) all processes are trying to kill each other

d) None

Question 132

A process stack does not contain _____

a) Function parameters

b) Local variables

c) Return addresses

d) PID of child process

Question 133

CPU scheduling is the basis of _____

a) multiprogramming operating systems

b) larger memory sized systems

c) multiprocessor systems

d) none

Question 134

Which system call can be used by a parent process to determine the termination of child process?

a) wait

b) exit

c) fork

d) get

Question 135

The address of the next instruction to be executed by the current process is provided by the _____

a) CPU registers

b) Program counter

c) Process stack

d) Pipe

Question 136

A Process Control Block(PCB) does not contain which of the following?

a) Code

b) Stack

c) Bootstrap Program

d) Data

Question 137

The number of processes completed per unit time is known as _____

a) Output

b) Throughput

c) Efficiency

d) Capacity

Question 138

The state of a process is defined by _____

a) the current activity of the process

b) the final activity of the process

c) the activity just executed by the process

d) the activity to next be executed by the process

Question 139

Which of the following is not the state of a process?

a) New

b) Old

c) Waiting

d) Running

Question 140

What is a Process Control Block?

a) Process type variable

b) Data Structure

c) A secondary storage section

d) A Block in memory

Question 141

The entry of all the PCBs of the current processes is in _____

a) Process Register

b) Program Counter

c) Process Table

d) Process Unit

Question 142

What is the degree of multiprogramming?

a) the number of processes executed per unit time

b) the number of processes in the ready queue

c) the number of processes in the I/O queue

d) the number of processes in memory

Question 143

A single thread of control allows the process to perform _____

a) only one task at a time

b) multiple tasks at a time

c) only two tasks at a time

d) all

Question 144

Which one of the following is not true?

a) kernel remains in the memory during the entire computer session

b) kernel is made of various modules which can not be loaded in running operating system

c) kernel is the first part of the operating system to load into memory during booting

d) kernel is the program that constitutes the central core of the operating system

Question 145

What is the objective of multiprogramming?

a) Have a process running at all time

b) Have multiple programs waiting in a queue ready to run

c) To increase CPU utilization

d) None

Question 146

Which of the following do not belong to queues for processes?

a) Job Queue

b) PCB Queue

c) Device Queue

d) Ready Queue

Question 147

When the process issues an I/O request _____

a) It is placed in an I/O queue

b) It is placed in a waiting queue

c) It is placed in the ready queue

d) It is placed in the Job queue

Question 148

What will happen when a process terminates?

a) It is removed from all queues

b) It is removed from all, but the job queue

c) Its process control block is de-allocated

d) Its process control block is never de-allocated

Question 149

What is a long-term scheduler?

a) It selects processes which have to be brought into the ready queue

b) It selects processes which have to be executed next and allocates CPU

c) It selects processes which heave to remove from memory by swapping

d) None

Question 150

If all processes I/O bound, the ready queue will almost always be _____ and the Short term Scheduler will have a _____ to do.

a) full, little

b) full, lot

c) empty, little

d) empty, lot

Question 151

What is a medium-term scheduler?

a) It selects which process has to be brought into the ready queue

b) It selects which process has to be executed next and allocates CPU

c) It selects which process to remove from memory by swapping

d) None

Question 152

What is a short-term scheduler?

a) It selects which process has to be brought into the ready queue

b) It selects which process has to be executed next and allocates CPU

c) It selects which process to remove from memory by swapping

d) None

Question 153

The primary distinction between the short term scheduler and the long term scheduler is _____

a) The length of their queues

b) The type of processes they schedule

c) The frequency of their execution

d) None

Question 154

The only state transition that is initiated by the user process itself is _____

a) block

b) wakeup

c) dispatch

d) none

Question 155

Which one of the following errors will be handle by the operating system?

a) lack of paper in printer

b) connection failure in the network

c) power failure

d) all

Question 156

In a time-sharing operating system, when the time slot given to a process is completed, the process goes from the running state to the _____

a) Blocked state

b) Ready state

c) Suspended state

d) Suspended state

Question 157

In a multiprogramming environment _____

a) the processor executes more than one process at a time

b) the programs are developed by more than one person

c) more than one process resides in the memory

d) a single user can execute many programs at the same time

Question 158

Suppose that a process is in “Blocked” state waiting for some I/O service. When the service is completed, it goes to the _____

a) Running state

b) Ready state

c) Suspended state

d) Terminated state

Question 159

The context of a process in the PCB of a process does not contain _____

a) the value of the CPU registers

b) the process state

c) memory-management information

d) context switch time

Question 160

Which of the following need not necessarily be saved on a context switch between processes?

a) General purpose registers

b) Translation lookaside buffer

c) Program counter

d) All

Question 161

Which of the following does not interrupt a running process?

a) A device

b) Timer

c) Schedular process

d) Power Failure

Question 162

Which process can be affected by other processes executing in the system?

a) cooperating process

b) child process

c) parent process

d) init process

Question 163

When several processes access the same data concurrently and the outcome of the execution depends on the particular order in which the access takes place is called _____

a) dynamic condition

b) race condition

c) essential condition

d) critical condition

Question 164

If a process is executing in its critical section, then no other processes can be executing in their critical section. What is this condition called?

a) mutual exclusion

b) critical exclusion

c) synchronous exclusion

d) asynchronous exclusion

Question 165

Which one of the following is a synchronization tool?

a) thread

b) pipe

c) semaphore

d) socket

Question 166

Where is the operating system placed in the memory?

a) either low or high memory (depending on the location of interrupt vector)

b) in the low memory

c) in the high memory

d) none

Question 167

A semaphore is a shared integer variable _____

a) that can not drop below zero

b) that can not be more than zero

c) that can not drop below one

d) that can not be more than one

Question 168

Mutual exclusion can be provided by the _____

a) mutex locks

b) binary semaphores

c) both mutex locks and binary semaphores

d) none

Question 169

When high priority task is indirectly preempted by medium priority task effectively inverting the relative priority of the two tasks, the scenario is called _____

a) priority inversion

b) priority removal

c) priority exchange

d) priority modification

Question 170

Process synchronization can be done on _____

a) hardware level

b) software level

c) both hardware and software level

d) none

Question 171

A monitor is a module that encapsulates _____

a) shared data structures

b) procedures that operate on shared data structure

c) synchronization between concurrent procedure invocation

d) all

Question 172

To enable a process to wait within the monitor _____

a) a condition variable must be declared as condition

b) condition variables must be used as boolean objects

c) semaphore must be used

d) all

Question 173

Restricting the child process to a subset of the parent's resources prevents any process from _____

a) overloading the system by using a lot of secondary storage

b) under-loading the system by very less CPU utilization

c) overloading the system by creating a lot of sub-processes

d) crashing the system by utilizing multiple resources

Question 174

A parent process calling _____ system call will be suspended until children processes terminate.

a) wait

b) fork

c) exit

d) exec

Question 175

Cascading termination refers to termination of all child processes if the parent process terminates _____

a) Normally

b) Abnormally

c) Any one

d) None

Question 176

With _____ only one process can execute at a time; meanwhile all other process are waiting for the processor. With _____ more than one process can be running simultaneously each on a different processor.

a) Uniprogramming, Multiprogramming

b) Multiprocessing, Multiprogramming

c) Multiprogramming, Multiprocessing

d) Uniprogramming, Multiprocessing

Question 177

If a process fails, most operating system write the error information to a _____

a) new file

b) another running process

c) log file

d) none

Question 178

In UNIX, the return value for the fork system call is _____ for the child process and _____ for the parent process.

a) A Negative integer, Zero

b) Zero, A Negative integer

c) Zero, A nonzero integer

d) A nonzero integer, Zero

Question 179

The child process can _____

a) be a duplicate of the parent process

b) never be a duplicate of the parent process

c) cannot have another program loaded into it

d) never have another program loaded into it

Question 180

The child process completes execution, but the parent keeps executing, then the child process is known as _____

a) Orphan

b) Zombie

c) Body

d) Dead

Question 181

What is Interprocess communication?

a) allows processes to communicate and synchronize their actions when using the same address space

b) allows processes to communicate and synchronize their actions

c) allows the processes to only synchronize their actions without communication

d) none

Question 182

Message passing system allows processes to _____

a) communicate with each other without sharing the same address space

b) communicate with one another by resorting to shared data

c) share data

d) name the recipient or sender of the message

Question 183

Which of the following two operations are provided by the IPC facility?

a) write & delete message

b) delete & receive message

c) send & delete message

d) receive & send message

Question 184

Messages sent by a process _____

a) have to be of a fixed size

b) have to be a variable size

c) can be fixed or variable sized

d) none

Question 185

The link between two processes P and Q to send and receive messages is called _____

a) communication link

b) message-passing link

c) synchronization link

d) all

Question 186

Which of the following are TRUE for direct communication?

a) A communication link can be associated with N number of process(N = max. number of processes supported by system)

b) A communication link is associated with exactly two processes

c) Exactly N/2 links exist between each pair of processes(N = max. number of processes supported by system)

d) Exactly two link exists between each pair of processes

Question 187

In indirect communication between processes P and Q _____

a) there is another process R to handle and pass on the messages between P and Q

b) there is another machine between the two processes to help communication

c) there is a mailbox to help communication between P and Q

d) none

TOC

Question 1

Statement 1: A Finite automata can be represented graphically;
 Statement 2: The nodes can be its states;
 Statement 3: The edges or arcs can be used for transitions
 Which of the following make the correct combination?

a) Statement 1 is false but Statement 2 and 3 are correct

b) Statement 1 and 2 are correct while 3 is wrong

c) None of the mentioned statements are correct

d) All of the mentioned statements are correct

Question 2

In a Turing machine the header can move over two cells at the same time

a) true

b) false

c) maybe false in some cases

Question 3

John is asked to make an deterministic finite automaton which accepts a given string for all the occurrence of ‘1001’ in it. How many number of transitions would John use such that, the string processing application works?

a) 10

b) 11

c) 12

d) 15

Question 4

The production of form -> ε is called:

a) Sigma

b) Null Transition

c) Null state

d) All of the mentioned

Question 5

The number of tuples in an Non Deterministic Finite Automaton:

a) 5

b) 6

c) 7

d) 4

Question 6

Predict the number of transitions required to build a DFA for the following language using only 3 states: ∑{0,1}
 L= {w | w ends with 00}

a) 6

b) 3

c) 4

d) Cannot be said

Question 7

In NFA a state cannot travel to more than one state using the same input symbol

a) true

b) false

Question 8

In the tuples which is used to represent an automata Q represents the states in the automata

a) true

b) false

Question 9

In the tuples which is used to represent an automata, F represents the non final state elements in the automata

a) true

b) false

Question 10

All DFA's can be called as NFA's

a) true

b) false

Question 11

Kleen closure property is represented with * symbol

a) true

b) false

Question 12

Alphabet set is formed using symbols

a) True

b) False

c) May not be true in all cases

d) May be false in some cases

Question 13

Positive closure represented with *

a) true

b) false

Question 14

In a Context Free Grammar every Terminal produces a language

a) true

b) false

Question 15

The formal notation of a context free grammar is

a) G={V,T,P,S}

b) G={V,T,P}

c) G={V,T,S}

d) G={G1,V,T,P,S}

Question 16

A variable non terminal may be _____ if there is no way of getting terminal from it.

a) useful

b) useless

c) cannot determine

Question 17

which of the following grammar are useless productions
 S→aS | A | C
 A→a
 B→aa
 C→aCb

a) S→aS | A | C
 A→a
 B→aa
 C→aCb

b) S→aS | A | C
 A→a

c) S→aS | A | C
 B→aa

d) S→C
 B→aa
 C→aCb

Question 18

A grammar is said to be ambiguous when

a) if there exists some w∈L(G) which has atleast two left most or right most derivations

b) if there exists some w∈L(G) which has atleast one left most or right most derivations

c) if there exists some w∈L(G) which no left most or right most derivations

d) all

Question 19

Any production of a context free grammar of the form A→B where A,B∈V is called

a) Unit production

b) Useless Symbol

c) Null Production

d) All

Question 20

In G{V,T,P,S}
 where V represents Terminals

a) true

b) false

c) maybe true in some cases

Question 21

A grammar G is called a CFG if production has

a) only one terminal in LHS

b) any combination of terminals and non-terminals on its RHS

c) RHS always starting with non-terminals

d) only one terminal in LHS and any combination of terminals and non-terminals on its RHS

Question 22

Which of the following grammar is ambiguous

a) S→aS|Sa|a

b) S→0S0|1S1|0|1|ε

c) S→S0S0S|1

d) All of these

Question 23

Languages are conditions in forming strings over the symbols available in the alphabet set

a) true

b) false

c) may be true in some cases

d) may be false in most of the cases

Question 24

The CFG for the language L={0^n 1^n |n>=1} is

a) S→0S|S0|1

b) S→0S0|1S1|0|1|ε

c) S→S0S0S|1

d) S→0S1 | 01

Question 25

Which of the following machines accepts regular languages

a) NFA

b) DFA

c) PDA

d) All

Question 26

Which of the following languages can be accepted by DPDA

a) L={WcW^r | W{a,b}*}

b) L={a^nb^n|n>=1}

c) both are correct

d) both are wrong

Question 27

which of the following languages are accepted by NPDA but not by DPDA

a) palindrome of even length

b) L={0^m 1^n 2^p | m,n,p>=0 and m=n}

c) L={0^m 1^n 2^m | m=n+p,n,p>=0}

d) none

Question 28

PDA is equivalent to finite automata except that it has an extra memory unit called as queue

a) true

b) false

Question 29

Finite state machines have limited memory only in the form of states.

a) true

b) false

Question 30

PDA may be deterministic or non-deterministic.

a) true

b) false

Question 31

In formal notation Z0 is stack bottom symbol

a) true

b) false

Question 32

which of the following language is not regular

a) L={a^n | n>=0}

b) L={a^n*n | n>=0}

c) L={a^2n | n>=0}

d) all

Question 33

If the PDA does not stop on an accepting state and the stack is not empty, the string is:

a) rejected

b) accepted

c) all of the mentioned

d) none of the mentioned

Question 34

Strings can have symbols which are not shown in the alphabet set

a) yes

b) no

c) maybe in special case

Question 35

The operations of PDA never work on elements, other than the top symbol in stack.

a) TRUE

b) FALSE

Question 36

Languages determine the nature of string formed using the given alphabet set

a) yes

b) no

c) maybe true in some cases

Question 37

Identify the string set for the $\Sigma\{0,1\}*$ even count of zeros and even count of ones

a) {00,11,0101,00001111,0101,.....}

b) {ε,00,11,0101,00001111,0101,.....}

c) {ε,010,101,001,0000111,0101,.....}

d) None of the choice

Question 38

Σ {0,1}Σ^3 = {000, 001, 010, 011, 100, 101, 110, 111 }

a) correct

b) wrong

Question 39

Identify the string set for the Σ {0,1}+ even count of zeros and odd count of ones

a) {ε,001,100,0000111,10101}

b) {001,100,01,0000111,10101}

c) {0101,1010,1001,01000111,101001}

d) {001,100,001,0000111,10101}

Question 40

In DFA, a state cannot travel to more than one state with the same input symbol

a) true

b) false

c) maybe true in some cases

CAO

Question 1

In memory transfer location address is supplied by____ that puts this on address bus

a) ALU

b) CPU

c) MAR

d) MDR

Question 2

Control memory is part of _____ that has addressable storage registers and used as temporary storage for data

a) ROM

b) RAM

c) CPU

d) Memory

Question 3

How many modes the address in control memory are divided into?

a) 2

b) 3

c) 5

d) 7

Question 4

which of the following is interrupt mode

a) Task mode

b) Executive mode

c) Both a & b

d) None of these

Question 5

Mode of addresses in control memory are:

a) Executive mode

b) Task mode

c) Both Executive mode & Task Mode

d) None of these

Question 6

On what method search in cache memory used by the system

a) Cache directing

b) Cache mapping

c) Cache controlling

d) Cache invalidation

Question 7

Microprograms are usually stored in

a) ROM

b) RAM

c) SAM

d) SAN

Question 8

In which addressing the operand is actually present in instruction

a) Immediate addressing

b) Direct addressing

c) Register addressing

d) None of these

Question 9

In which addressing the simplest addressing mode where an operand is fetched from memory is_____

a) Immediate addressing

b) Direct addressing

c) Register addressing

d) None of these

Question 10

which addressing is a way of direct addressing

a) Immediate addressing

b) Direct addressing

c) Register addressing

d) None of these

Question 11

In which mode the main memory location holds the Effective Address of the operand

a) Immediate addressing

b) Direct addressing

c) Register addressing

d) Indirect addressing

Question 12

Operation of memory transfer are

a) Read

b) Write

c) Both Read and Write

d) None

Question 13

Which addressing is an extremely influential way of addressing

a) Displacement addressing

b) Immediate addressing

c) Direct addressing

d) Register addressing

Question 14

Which algorithm is a multiplication algorithm which multiplies two signed binary numbers in 2's complement notation

a) Usual multiplication

b) Booth's multiplication

c) Both

d) None of these

Question 15

. _____interrupt establishes a priority over the various sources to determine which request should be entertained first

a) Priority interrupt

b) Polling

c) Daisy chaining

d) None of these

Question 16

An instruction code must specify the address of the____

a) Opcode

b) Operand

c) Both

d) None

Question 17

In memory, the read operation puts memory address on to a register known as

a) PC

b) ALU

c) MAR

d) All of these

Question 18

Which operation puts memory address in memory address register and data in DR

a) Memory read

b) Memory write

c) Both

d) None

Question 19

The memory bus is also referred as_____

a) Data bus

b) Address bus

c) Memory bus

d) All of these

Question 20

DDR2 stands for

a) Double data rate 2

b) Data double rate 2

c) Dynamic data rate 2

d) Dynamic double rate 2

Question 21

SDRAM stands for

a) System dynamic random access memory

b) Synchronous dynamic random access memory

c) Both

d) None

Question 22

_____is the sequence of operations performed by CPU in processing an instruction

a) Execute cycle

b) Fetch cycle

c) Decode

d) Instruction cycle

Question 23

IRET stand for

a) Interrupt enter

b) Interrupt return

c) Interrupt delete

d) None of these

DBMS

Question 1

While drawing the ER diagram for a relation “student registers a course”, “registers” should be mentioned _____ 2 rectangles.

a) inside a diamond between

b) over the line connecting

c) inside an oval between

d) inside a rectangle between

Question 2

What is the number of candidate keys for a relation R(A,B,C,D) having functional dependencies {AB->CD, D->B, C->A}?

a) 2

b) 3

c) 4

d) 5

Question 3

In a relation R(A,B,C,D) having functional dependencies {AB->CD, D->B}, the extraneous attribute in AB->CD is

a) A

b) B

c) C

d) None

Question 4

The highest normal form of a relation R(A,B,C,D) with functional dependencies {A->BCD, BC->AD, D->B} is

a) 1NF

b) 2NF

c) 3NF

d) BCNF

Question 5

The non-prime attributes of the relation R(A,B,C,D,E) with functional dependencies {AE->BC, AC->D, CD->BE, D->E} is/are:

a) A,C,D,E

b) B

c) A,C

d) B,D,E

Question 6

In the relation R(A,B,C,D,E,F) with functional dependencies {AB->C, C->DE, E->F, D->A, C->B}, AB+ is

a) {C}

b) {C,D,E}

c) {C,D,E,F}

d) {A,B,C,D,E,F}

Question 7

What is the relational algebra expression corresponding to the SQL query “SELECT city FROM store”?

a) sigmacity(store)

b) picity(store)

c) sigmastore(city)

d) pistore(city)

Question 8

Which of the following expressions is used to fetch the details of all stores located in India?

a) sigmacountry=’India’(store)

b) picountry=’India’(store)

c) sigmastore(country=’India’)

d) pistore(country=’India’)

Question 9

Which of the following is not a valid schedule of 2 transactions, T1: {R(X), W(X)} and T2: {R(Y), W(Y)}?

a) R(X), W(X), R(Y), W(Y)

b) R(Y), W(Y), R(X), W(X)

c) W(X), W(Y), R(X), R(Y)

d) R(X), R(Y), W(X), W(Y)

Question 10

Which of the following pairs of operations conflict in the transactions, T1: {R1(X), W1(X)} and T2: {R2(X), W2(Y)}

a) R1(X),W1(X)

b) R1(X),R2(X)

c) W1(X),R2(X)

d) R1(X),W2(Y)

Question 11

Which of the following is a conflict serializable schedule of the transactions, T1: {R1(X), W1(X), R1(Y)} and T2: {R2(Y), R2(X), W2(X)}?

a) R1(X), R2(Y), R2(X), W1(X), W2(X), R1(Y)

b) R2(Y), R2(X), R1(X), W1(X), W2(X), R1(Y)

c) R1(X), W1(X), R2(Y), R2(X), W2(X), R1(Y)

d) R2(Y), R2(X), R1(X), W2(X), W1(X), R1(Y)

Question 12

A database designed to store global information contains a field “contact number” in a specific relation. If contact number is to be saved in multiple parts like country code and the actual number, what kind of attribute should be used to represent contact number in the ER diagram?

a) Atomic attribute

b) Composite attribute

c) Multi-valued attribute

d) Derived attribute

Question 13

Which of the following statement(s) is/are TRUE?
 Consider the following two statements about database transaction schedules:
 I. Strict two-phase locking protocol produces conflict serializable schedules which are not recoverable.
 II. Timestamp-ordering concurrency control protocol with Thomas' Write Rule can produce view serializable schedules that are not conflict serializable.

a) I only

b) II only

c) Both I and II

d) Neither I nor II

Question 14

Which of the following is not a goal of concurrent execution of a transaction?

a) Improved Resource Utilization

b) Improved Throughput

c) Reduced probability of deadlock

d) Reduced waiting time

Question 15

Which one of the following is true about the concurrency?

a) It ensures that no interference between simultaneous operations on the database

b) It ensures that a transaction allows either all of its operations are executed or none

c) It make sure that database transactions allows any integrity violation in the database.

d) None of the options

Question 16

Which of the following will not allow both dirty reads and non-repeatable reads, but allows phantom reads?

a) Read committed

b) Read uncommitted

c) Repeatable read

d) Serializable

Question 17

Which of the following is false?

a) Ensuring durability is the responsibility of recovery management component

b) Ensuring isolation is the responsibility of concurrency management

c) Ensuring atomicity or consistency is the responsibility of recovery management component

d) None of the options

Question 18

Which of the following is true?

a) All the schedules which are allowed under Thomas write rule are also allowed under basic timestamp protocol

b) All the schedules which are allowed under basic timestamp rule are also allowed under multi-version timestamp protocol

c) All the schedules which are allowed under multi-version timestamp protocol are also allowed under Thomas write rule

d) All are true

Question 19

Which of the following statement is not correct for serializable of transactions?

a) In a serial schedule, each transaction is independent of others.

b) Every schedule is serializable

c) In non-serial schedule may not always result in an incorrect outcome

d) In non-serial schedule, may result with overlapping of two transaction's executions

Question 20

Which of the following is not a recovery technique?

a) Deferred update

b) Immediate update

c) Two-phase commit

d) Recovery management

Question 21

Rollback of transactions is normally used to :

a) Recover from transaction failure

b) Update the transaction

c) Retrieve old records

d) Repeat a transaction

Question 22

Which of the following are true about indexes?

a) Indexes will improve the performance of the database table, even it is updated frequently

b) Indexes makes sql server engine’s task even tougher, especially with large keys

c) Both

d) None of the options

Question 23

What kind of a relation is “a student registers for more than one course for a semester”?

a) one-to-many

b) many-to-many

c) one-to-one

d) None

Question 24

Consider a B+ tree, the size of the block and node is same as 4 kilobytes. Suppose each index entry is 40 bytes long, find out the height of the tree of a file with one million search key values?

a) 5

b) 4

c) 6

d) 3

Question 25

The schedule S: r1(X), w3(X), w3(Y), w2(Y), r1(Z), w3(Z), w2(Z) is

a) Both view serializable and conflict serializable schedule

b) View serializable but not conflict serializable schedule

c) Only View serializable

d) Only conflict serializable

Question 26

Which of the following statement(s) is/are not TRUE about the B+ tree, which is used for creating an index of a database table?

a) B+ Tree is a height-balanced tree

b) Non-leaf nodes have pointers to data records

c) Key values in each node are kept in sorted order

d) Each leaf node has a pointer to the next leaf node

Question 27

B+ Trees are normally considered as BALANCED, since

a) Lengths of the paths from the root to all leaf nodes are all equal.

b) Lengths of the paths from the root to all leaf nodes differ from each other by at most

c) Number of children of any two non-leaf sibling nodes differ by at most 1.

d) Number of records in any two leaf nodes differ by at most 1

Question 28

Primary indexes of multilevel indexes are created usually based on its first level and it is normally referred as

a) zero level of multilevel index

b) third level of multilevel index

c) second level of multilevel index

d) first level of multilevel index

Question 29

Which of the following is/are correct?

a) B-trees are for storing data on disk and B+ trees are for main memory.

b) Range queries are faster on B+ trees.

c) B-trees are for primary indexes and B+ trees are for secondary indexes.

d) The height of a B+ tree is independent of the number of records.

Question 30

Find the maximum and minimum number of keys in the internal nodes of B – Tree if the order of the node is four,

a) 4,2

b) 3,1

c) 4,0

d) 4,3

Question 31

Which of the following statement(s) are incorrect

a) In non-relational databases, schemas are need to be defined before adding data

b) NoSQL databases allows the insertion of data without a predefined schema

c) NewSQL databases allows the insertion of data without a predefined schema

d) All the statements

Question 32

NoSQL databases is used mainly for handling large volumes of _____ data.

a) Unstructured

b) Structured

c) Semi-structured

d) All the options

Question 33

Which of the following are the NoSQL databases?

a) Key-value

b) Wide-column

c) Document

d) All the options

Question 34

If an athlete can participate in a maximum of 3 activities, what is cardinality ratio of ATHLETE:ACTIVITY in PARITICIPATE?

a) 1:n

b) 2:1

c) 1:3

d) n:1

Question 35

Consider the following schedule and state that which of the following statement(s) is (are) true?
 S : r1(x),w1(x),r2(x),r1(y),w1(y),w2(x),r2(y), c1, c2;
 I: S is recoverable.
 II: If the order of w1(x) and w2(x) are interchanged, then S will be a non-recoverable schedule.

a) Only I

b) Only II

c) Both I and II

d) None of the options

Question 36

For the given schedule, which of the following statements is true about the schedule S?
 S : r1(y), r2(x), r2(y), r3(y), w2(x), w1(y), w3(x), r1(x)

a) S is the conflict serializable but not view serializable

b) S is neither conflict serializable nor view serializable

c) S is both conflict serializable as well as view serializable

d) S is view serializable but not conflict serializable

Question 37

Which of the following is not an entity set in a SHOPPING mini world?

a) Customers

b) Stock

c) Products

d) ProductName

Question 38

Identify the type of database system required to store a research document.

a) Multimedia Database

b) Geographical Information System

c) Data Warehouse

d) Textual Database

Question 39

In a Bank scenario, the window visible to a customer while doing transactions via net banking is referred to as -----

a) The database state

b) The Schema

c) The View

d) Data dictionary

Question 40

A bank operation accommodates several operations in a day. A snapshot taken at 4 p.m to know the transactions currently happening is referred to as----------

a) Database Schema

b) Database state

c) Abstraction

d) Catalogue

Question 41

Which of the following is a possible schema construct in a supermarket mini world?

a) Products

b) inventory

c) sales

d) All the options

Question 42

Suppose you are assigned to do a standalone database application, what type of DBMS architecture is required?

a) Single tier architecture

b) Two tier

c) Three tier

d) None of the mentioned

Question 43

Identify the possible entity sets of a Cafeteria Data base.

a) Customers, menu manager, meal Delivery

b) Customers, menuManager, itemsProcurement, mealDelivery,billGeneration

c) Customers, menu manager, Bill generation

d) FoodMenu

Question 44

The attribute Address could be structured as door no, street_name, City and Zipcode. This type of attribute is called ---------

a) Simple attribute

b) Multivalued attribute

c) Composite attribute

d) Derived attribute

Question 45

For a relation R(A,B,C,D,E,F,G), the candidate keys are AB, AC, and DE. Which of the following is not a proper subset of the candidate key?

a) BC

b) DE

c) C

d) AD

Question 46

The function that an entity Instructor plays in a relationship teaches refers to the -------

a) Participation

b) Role

c) Instance

d) Position

Question 47

Which of the following can be a multivalued attribute?

a) Phone_number

b) Name

c) Date_of_birth

d) All the options

Question 48

Identify the key attribute of the entity PROJECT(ProjectId, Projectname, ProjLocation)

a) ProjectId, ProjectName

b) ProjectName

c) ProjLocation

d) ProjectId

Question 49

The attribute experience is calculated from the attribute dateOfJoining. What is the type of the attribute?

a) Derived attribute

b) Descriptive attribute

c) Simple attribute

d) Multivalued attribute

Question 50

SLOT is a weak entity of the University schema. Which of the statements is not true?

a) It requires a identifying relationship TAKES

b) It has a identifying owner entity TIMETABLE

c) It should have total participation with respect to the entity TAKES

d) There is no existence dependency of the entity SLOT with respect to the entity TIMETABLE

Question 51

ORDERS is an entity set with attributes (orderId, items, price) and SHIPPING entity is a specialized version of the ORDERS entity. Which of the following are the attribute set of SHIPPING entity?

a) (OrderId, items, price)

b) (Orderid, Shippingid)

c) (OrderId, items, price, ShippingId,FromLocation,ToLocation)

d) (OrderId, Items, ShippingId,FromLocation,ToLocation)

Question 52

If an attribute is added to the superclass entity, the condition for propagating this change in subclass entity is

a) Attribute defined membership

b) Predicate defined membership

c) Both Attribute defined and Predicate defined membership

d) Overlapping participation

Question 53

“Customer buys Products”. Identify the relationship set.

a) Customer

b) Products

c) Buys

d) Customer and Products

Question 54

The entity set MovieZone is classified as Hollywood and Bollywood. This process is called ------

a) Generalization

b) Specialization

c) Inheritance

d) Constraint generalization

Question 55

Identify the simple attribute.

a) name

b) grade

c) Color

d) VehicleRegistrationNumber

Question 56

If a relational schema "R" is in BCNF, which of the following statements is true?

a) R is in 4NF

b) R is in 2NF but not in 3NF

c) R is not in 1NF

d) R is in 2NF and 3NF

Question 57

An entity set DOCTOR has a total participation on the relationship DIANOSES that connects it to the entity set PATIENTS. What would be the minimum cardinality constraint on the relationship?

a) 1:1

b) M:N

c) 1:N

d) N:1

Question 58

An entity set SHIPPING has the attributes (shippingId, ShipLocations). Which of the following is true?

a) ShippingId is the primary key

b) ShipLocations is the primary key

c) ShippingId, ShipLocations is a composite key

d) None of the options

Question 59

SENSE, SELECT, SCOPE are the entity sets with common characteristics. Which process can define an entity set called DEPARTMENTS?

a) Specialization

b) Generalization

c) Top down approach

d) Inheritance

Question 60

In a share market application, an entity set INVESTOR can buy or sell SHARES. This refers to ----------

a) Relationship

b) Role definition

c) Recursive relationship

d) Participation

Question 61

Which one of the following events in a database system can lead to an irrecoverable mistake?

a) A transaction reads a data item after it is written by an uncommitted transaction

b) A transaction reads a data item after it is written by a committed transaction

c) A transaction reads a data item after it is read by an uncommitted transaction

d) A transaction writes a data item after it is read by an uncommitted transaction

Question 62

Consider the table R with attributes A, B and C. The functional dependencies that hold on R are : A → B, C → AB. Which of the following statements is/are True ? I. The decomposition of R into R1(C, A) and R2(A, B) is lossless. II. The decomposition of R into R1(A, B) and R2(B, C) is lossy.

a) Only I

b) Only II

c) Both I and II

d) Neither I nor II

Question 63

Which of the following is correct with respect to Two phase commit protocol?

a) Ensures serializability

b) Prevents Deadlock

c) Detects Deadlock

d) Recover from Deadlock

Question 64

Consider a B+-tree in which the maximum number of keys in a node is 7. What is the minimum number of keys in any non-root node?

a) 3

b) 1

c) 2

d) 4

Question 65

A file is structured in such a way that the ordering of data records in certain indexes is the same as or similar to the ordering of data entries. That index is then called

a) Clustered

b) Sparse

c) Dense

d) Unclustered

Question 66

The physical position of a record is calculated by a formula that translates a file key into a position for the record is

a) B-Tree file

b) Hashed file

c) Indexed file

d) Sequential file

Question 67

SELECT _____ destination FROM flight;
 Fill the blank in the above SQL query with an appropriate keyword to display all non-repeating values of the column “destination” only once and values that occur once as such.

a) DIFFERENT

b) DISTINCT

c) ALL

d) UNIQUE

Question 68

Relation R has six attributes ABCDEF. Fields of R contain only atomic values. F = { A -> BC, B -> CFH, E -> A, F -> E} is a set of functional dependencies (FDs) so that F+ is exactly the set of FDs that hold for R. How many candidate keys does the relation R have?

a) 5

b) 6

c) 3

d) 4

Question 69

Consider the given functional dependencys (FDs) of a relation R has six attributes ABCDEF. Fields of R contain only atomic values. F = { A -> BC, B -> CFH, E -> A, F -> E} is a set of functional dependencies (FDs) so that F+ is exactly the set of FDs that hold for R. The relation R is in

a) 2NF, but not in 3NF.

b) 3NF, but not in BCNF.

c) 1NF, but not in 2NF.

d) BCNF.

Question 70

Consider the relation scheme R = {A, B, C, D, E, F, G} and the set of functional dependencies {{A, B} -> {C}, {B} -> {E, F}, {A, D} ->{G}} on R. What is the key for R?

a) {A, B, C, D, G}

b) {A, D}

c) {A, B}

d) {A, B, D}

Question 71

The maximum number of superkeys for the relation schema R(A, B, C, D, E, F, G, H) with H as the key is

a) 64

b) 128

c) 7

d) 256

Question 72

Select the 'False' statement from the following statements about Normal Forms:

a) Lossless preserving decomposition into 3NF is always possible

b) Any Relation with two attributes is in BCNF

c) Lossless preserving decomposition into BCNF is always possible

d) BCNF is stronger than 3NF

Question 73

A B-tree of order 4 is constructed by 8 successive insertions from scratch. What is the maximum number of operations for node splitting that can take place?

a) 3

b) 4

c) 5

d) 6

Question 74

The 'Durability' property among the ACID properties of a transaction requires that the modifications made by a successful transaction to the database survive.

a) Except in case of an Operating System crash

b) Always, even if there is a failure of any kind

c) Except in case of a power failure

d) Except in case of a Disk crash

Question 75

Consider the following schema:
 Emp (Empcode, Name, Sex, Salary, Deptt)
 A simple SQL query is executed as follows:
 SELECT Deptt FROM Emp
 WHERE sex = 'M'
 GROUP by Dept
 Having avg (Salary) > {select avg (Salary) from Emp}
 The output will be?

a) Average salary of male employee is more than the average salary of the organization

b) Average salary of male employee is equal to the average salary of the organization

c) Average salary of male employee is the average salary of the organization

d) Average salary of male employee is less than the average salary of the organization

Question 76

Consider the following query :
 SELECT E.eno, COUNT(*)
 FROM Employees E
 GROUP BY E.eno
 If an index on eno is available, the query can be answered by scanning only the index if

a) index can be hash or B+ tree and clustered or non-clustered

b) the index is only hash and clustered

c) the index is only B+tree and clustered

d) index can be hash or B+ tree and clustered

Question 77

Which level of locking provides the highest degree of concurrency in a relational data base?

a) Table

b) Row

c) Page

d) Page, table and row level locking allow the same degree of concurrency

Question 78

SELECT * FROM player WHERE num_of_matches>100 AND country='India';
 What fields will be outputted by the above SQL query?

a) player

b) num_of_matches

c) num_of_matches, country

d) All fields of player relation

Question 79

Which of the following SQL queries can be used to change "Mishra" to "Singh" in the "LastName" column in the People relation?

a) UPDATE People SET LastName='Mishra' INTO LastName='Singh'

b) UPDATE People SET LastName='Singh' WHERE LastName='Mishra'

c) MODIFY People SET LastName='Mishra' INTO LastName='Singh'

d) MODIFY People SET LastName='Singh' WHERE LastName='Mishra'

Web Technology – Comprehensive Examination – MCQs

1. Which company developed JavaScript?

A. Microsoft
B. Netscape
C. Google
D. Oracle

Answer: B. Netscape

2. Which symbol is used for single-line comments in JavaScript?

A. <!-- -->
B. //
C. ##
D. **

Answer: B. //

3. Which keyword declares a block-scoped variable?

A. var
B. let
C. static
D. function

Answer: B. let

4. How do you access a property name of an object student?

A. student(name)
B. student->name

C. student.name
D. student::name

Answer: C. student.name

5. Which method adds an element at the end of an array?

A. push()
B. pop()
C. shift()
D. unshift()

Answer: A. push()

6. Which method returns the length of a string?

A. size()
B. count()
C. length
D. strlen()

Answer: C. length

7. Which keyword stops loop execution?

A. stop
B. break
C. exit
D. return

Answer: B. break

8. Which operator checks both value and type?

A. ==
B. !=
C. ===
D. =

Answer: C. ===

9. What will be the output?

```
let str = "JavaScript";

console.log(str.substring(0,4));
```

A. Java
B. JavaS
C. Script
D. avaS

Answer: A. Java

10. Which method removes the last array element?

A. pop()
B. push()
C. shift()
D. delete()

Answer: A. pop()

11. What is the default value of an uninitialized variable?

A. null
B. 0
C. undefined
D. empty

Answer: C. undefined

12. Which method adds an element at the end of an array?

A. push()
B. pop()
C. shift()
D. unshift()

Answer: A. push()

13. Which object is used to work with dates and time in JavaScript?

A. Time
B. Calendar
C. Date
D. Clock

Answer: C. Date

14. Which method returns the current year from a Date object?

A. getYear()
B. getFullYear()
C. getDate()
D. getMonth()

Answer: B. getFullYear()

15. Which keyword defines a function?

A. method
B. function
C. def
D. fun

Answer: B. function

16. Which of the following is NOT a JavaScript data type?

A. Number
B. Boolean
C. Float
D. String

Answer: C. Float

17. Which method is used to search a string using a regular expression?

A. find()
B. search()
C. locate()
D. scan()

Answer: B. search()

18. What will be the output?

```
console.log(Array.isArray([10, 20, 30]));
```

A. false
B. true
C. undefined
D. Error

Answer: B. true

19. Which architecture separates presentation, application logic, and data storage?

A. Monolithic architecture
B. Client–Server architecture
C. Three-tier architecture
D. Peer-to-Peer architecture

Answer: C. Three-tier architecture

20. In web architecture, the browser acts as:

A. Server
B. Client
C. Middleware
D. Database

Answer: B. Client

21. Which protocol is primarily used for communication between web client and server?

A. FTP
B. SMTP
C. HTTP/HTTPS
D. TCP

Answer: C. HTTP/HTTPS

22. Which tier handles business logic in a three-tier web architecture?

A. Presentation tier
B. Data tier
C. Application tier
D. Network tier

Answer: C. Application tier

23. Which component stores and manages persistent data?

A. Web server
B. Application server
C. Client browser
D. Database server

Answer: D. Database server

24. What is the role of a web server (e.g., Apache, Nginx)?

A. Executes business logic
B. Stores user data
C. Handles HTTP requests and responses
D. Manages client UI

Answer: C. Handles HTTP requests and responses

25. Which architectural style is commonly used for scalable web services?

A. SOAP
B. REST
C. RPC
D. CGI

Answer: B. REST

26. Which component improves performance by storing frequently accessed data?

A. Load balancer
B. Cache
C. Firewall
D. Proxy

Answer: B. Cache

27. What is the main purpose of a load balancer?

A. Store data securely
B. Reduce server-side scripting
C. Distribute traffic across multiple servers
D. Convert HTTP to HTTPS

Answer: C. Distribute traffic across multiple servers

28. In MVC architecture, which component handles user input?

A. Model
B. View
C. Controller
D. Database

Answer: C. Controller

29. What is the main function of a web server?

A. Design web pages
B. Store databases
C. Handle HTTP requests and send responses
D. Execute client-side scripts

Answer: C. Handle HTTP requests and send responses

30. Which protocol is used for web communication?

A. FTP
B. SMTP

C. HTTP
D. SNMP

Answer: C. HTTP

31. Which HTTP method is used to retrieve data from a server?

A. POST
B. GET
C. PUT
D. DELETE

Answer: B. GET

32. Which HTTP method is mainly used to send data to the server?

A. GET
B. POST
C. TRACE
D. CONNECT

Answer: B. POST

33. Which component sends an HTTP request?

A. Web server
B. Database
C. Client/Browser
D. Application server

Answer: C. Client/Browser

34. What does HTTP stand for?

A. Hyper Transfer Text Protocol
B. HyperText Transfer Protocol
C. High Transfer Text Protocol
D. HyperText Transmission Process

Answer: B. HyperText Transfer Protocol

35. Which status code indicates “Page Not Found”?

A. 200
B. 301
C. 404
D. 500

Answer: C. 404

36 Which status code indicates a successful request?

A. 200
B. 404
C. 500
D. 403

Answer: A. 200

37. What is included in an HTTP response?

A. Only request header
B. Status code, headers, and body
C. Only URL
D. Only cookies

Answer: B. Status code, headers, and body

38. HTTPS differs from HTTP because it:

A. Uses FTP protocol
B. Encrypts data using SSL/TLS
C. Works only on local networks
D. Does not use servers

Answer: B. Encrypts data using SSL/TLS

39. What is the primary purpose of cookies?

A. Store server-side objects
B. Maintain user state on the client
C. Execute JavaScript code
D. Encrypt HTTP data

Answer: B. Maintain user state on the client

40. Where are cookies stored?

A. Web server
B. Database
C. Client browser
D. Application server

Answer: C. Client browser

41. Which cookie is deleted when the browser is closed?

A. Persistent cookie
B. Secure cookie
C. Session cookie
D. HTTP-only cookie

Answer: C. Session cookie

42. Sessions are mainly used to:

A. Store data permanently
B. Track users across multiple requests
C. Reduce server load
D. Improve page styling

Answer: B. Track users across multiple requests

43. Where is session data usually stored?

A. Client browser
B. Proxy server
C. Web server
D. DNS server

Answer: C. Web server

44. Which technique is used by AJAX to update a web page without reloading?

A. HTML forms
B. XML parsing
C. Asynchronous server communication
D. CSS styling

Answer: C. Asynchronous server communication

45. Which object is traditionally used in AJAX to send requests?

A. Fetch
B. XMLHttpRequest
C. Document
D. Window

Answer: B. XMLHttpRequest

46. What does AJAX stand for?

A. Asynchronous Java and XML
B. Asynchronous JavaScript and XML
C. Advanced JavaScript and XHTML
D. Asynchronous JSON and XML

Answer: B. Asynchronous JavaScript and XML

47. Which data format is most commonly used with modern AJAX?

A. XML
B. HTML
C. JSON
D. CSV

Answer: C. JSON

48. Which AJAX feature improves user experience?

A. Page reload after every request
B. Synchronous communication
C. Partial page updates
D. Server-side rendering only

Answer: C. Partial page updates

49. AJAX requests are typically handled in the background using:

A. Multithreading
B. Asynchronous calls
C. Batch processing
D. Compilation

Answer: B. Asynchronous calls

50. Which is more secure for storing sensitive user data?

A. Cookies
B. Sessions
C. Local storage
D. Cache

Answer: B. Sessions

51. AJAX allows:

A. Only synchronous data transfer
B. Page reload after every request
C. Communication with server without reloading the page
D. Only XML data transfer

Answer: C. Communication with server without reloading the page